

Graph Algorithms

Introduction and basic Definitions

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Why Networks?

**Networks are a general language
for describing complex systems**

Networks and Complex Systems

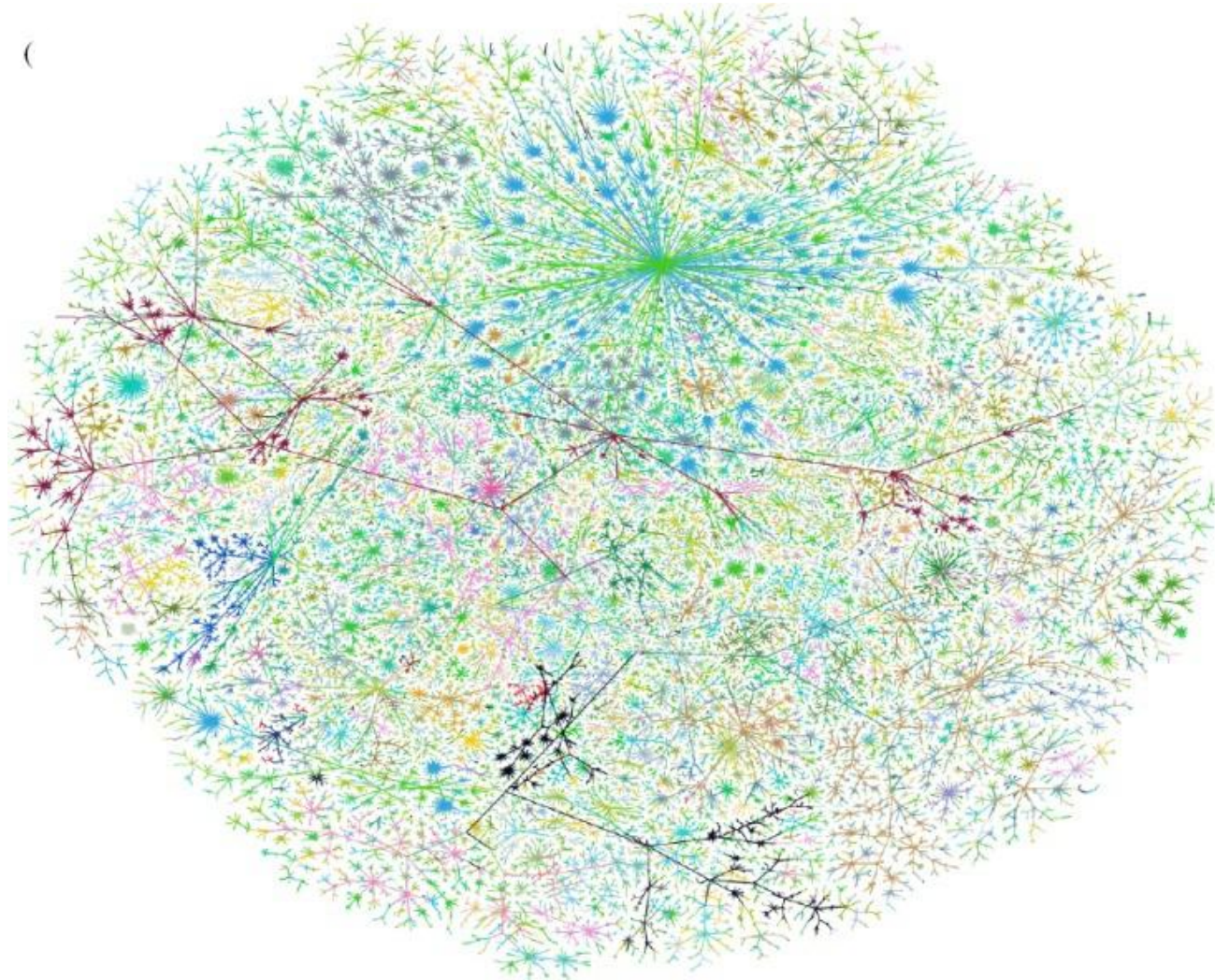
There are complex systems all around us:

- **Society** is a collection of 8+ billion individuals
- **Communication systems** link electronic devices
- **Information** and **knowledge** are organized and linked
- Interactions between thousands of **genes/proteins** regulate life
- Our **thoughts** are hidden in connections between billions of neurons

What do these systems have in common?

How can we represent them?

The Network



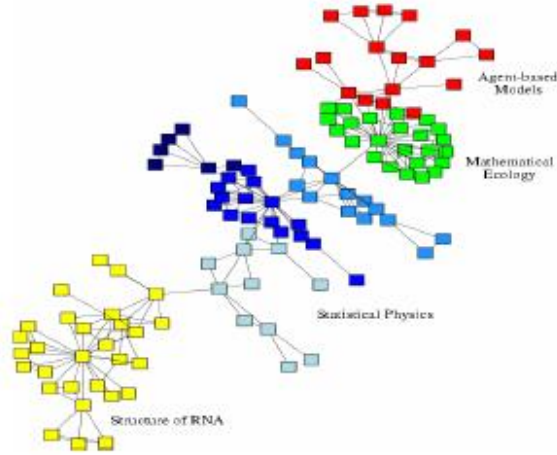
Behind many systems there is an intricate wiring diagram, **a network**, that defines the interactions between the components

We will never be able to model and predict these systems unless we understand the **networks** behind them!

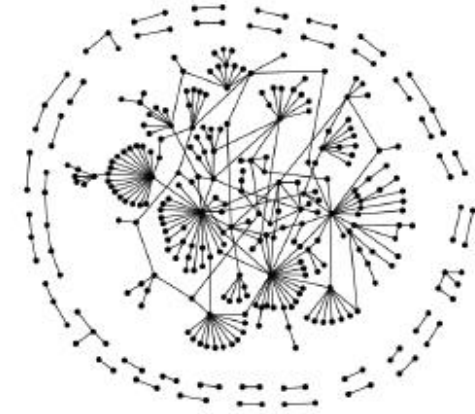
Many Types of Data are Networks



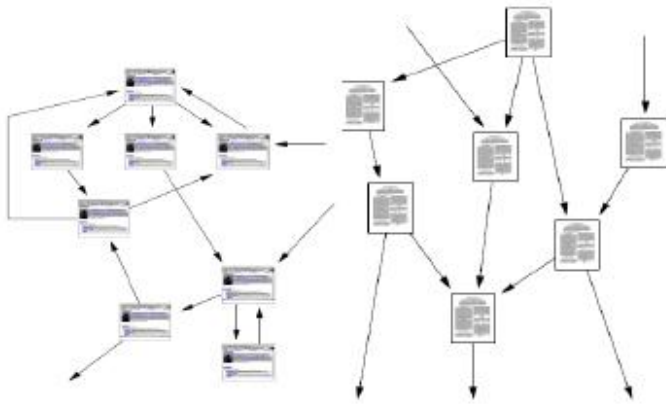
Social networks



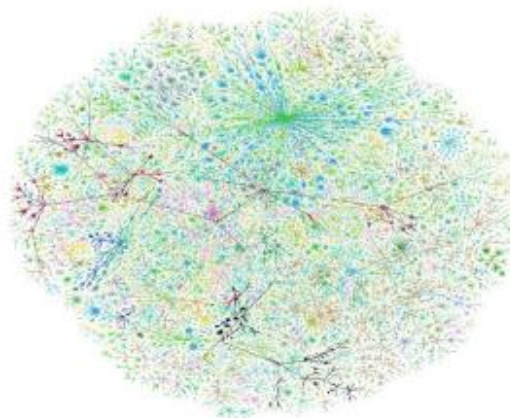
Economic networks



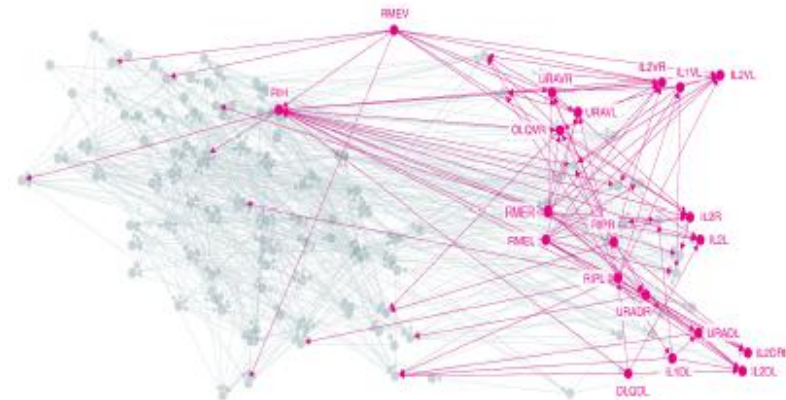
Communication graphs



Information networks:
Web & citations



Internet



Networks of neurons

Who cares about Networks ??

You Should !

Why Networks ? Now?

- **Universal language** for describing complex data
 - Networks from science, nature, and technology are more similar than one would expect
- **Shared vocabulary** between fields
 - CS, Social Science, Physics, Economics, Statistics, Biology
- **Data availability** & computational challenges
 - Web/mobile, bio, health, and medical, end more
- **Impact!**
 - Social networking, Social media, Drug design

Networks - Analysis

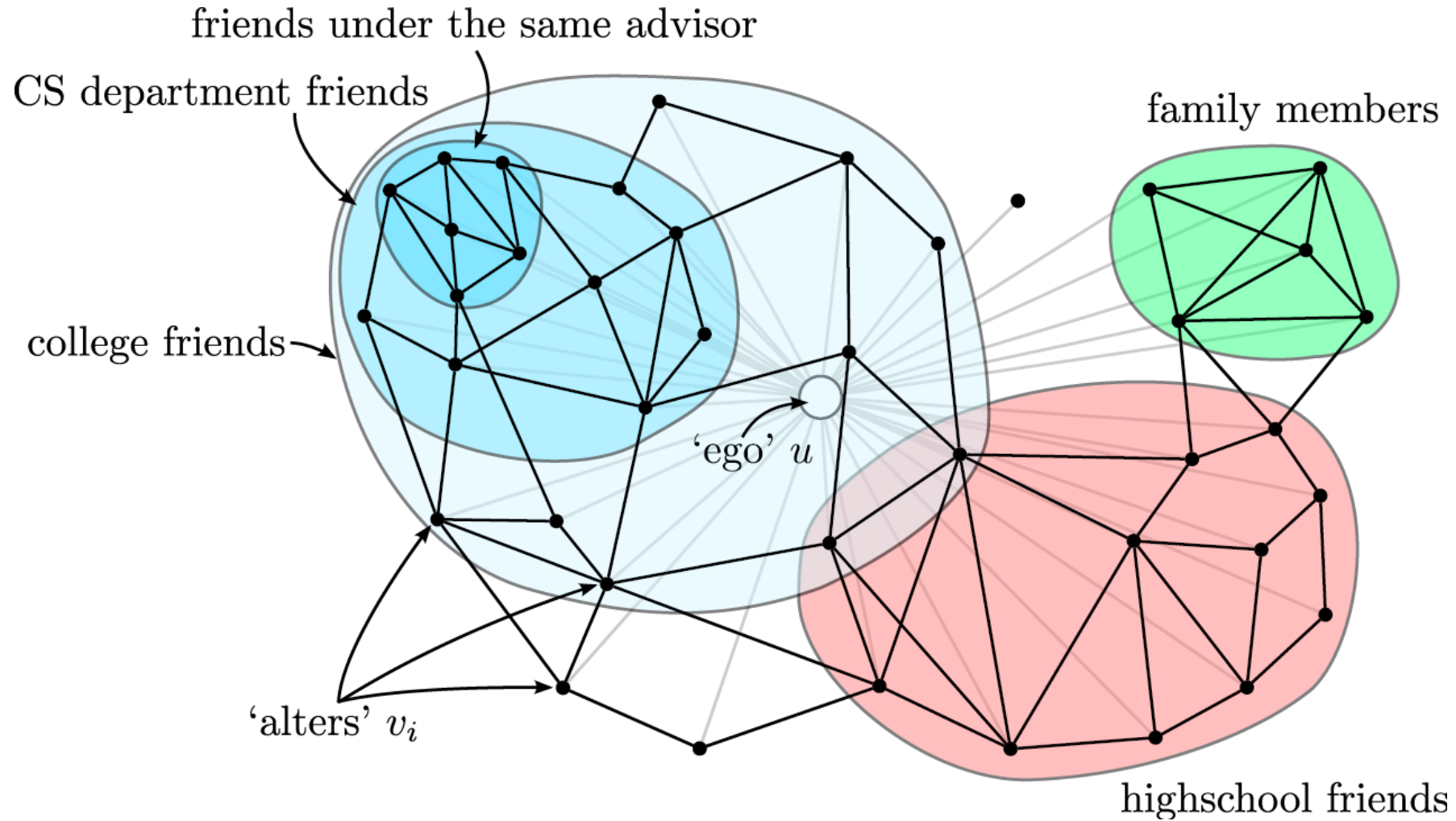
- Predict the type/color of a given node
 - Node classification
- Predict whether two nodes are linked
 - Link prediction
- Identify densely linked clusters of nodes
 - Community detection
- Measure similarity of two nodes/networks
 - Network similarity

Networks and their Applications

- Facebook / Meta [Social graph](#)



Networks - Applications

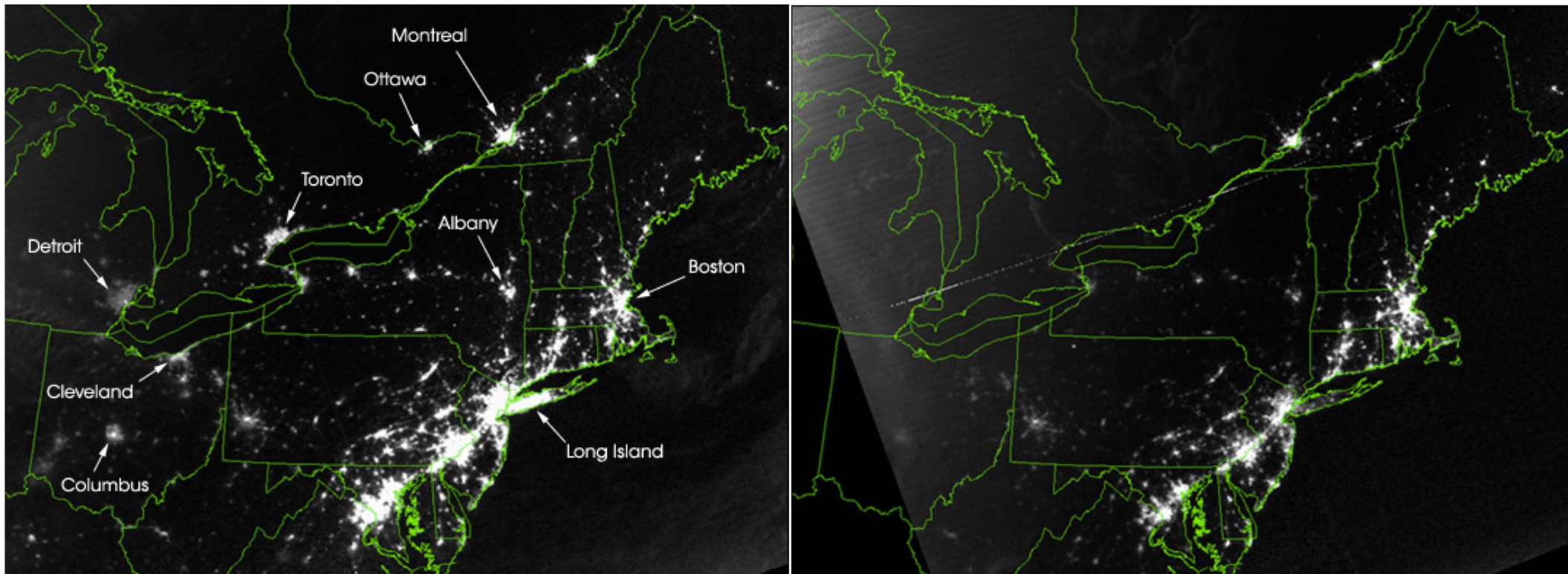


Discover Social Circles

Networks - Infrastructure

- **2003 Northeast Blackout - 55 million people**

- New York, New Jersey, Connecticut, Massachusetts, Michigan, Ontario (Canada)



20 hours before

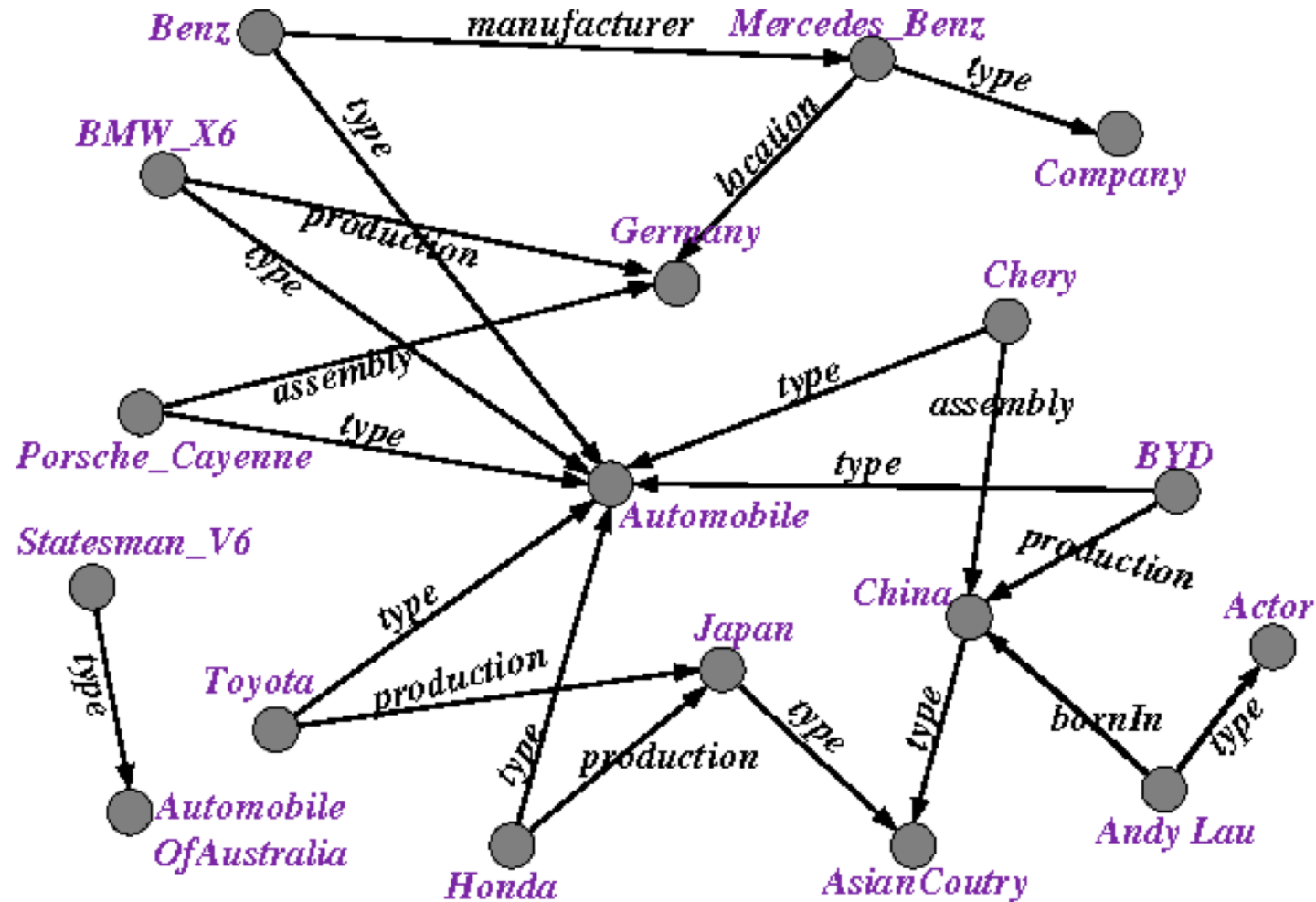
7 hours after

Networks - Infrastructure

2003 Blackout – Important Lessons

- We must understand how **network structure** affects the robustness of a system
- Develop quantitative tools to assess the interplay between network structure and the dynamical processes on the networks, and their impact on failures
- Failures follow reproducible laws, that can be quantified and even predicted using the tools of **networks**

Networks - Information and Knowledge



Knowledge Graph (Cars)

Networks - Information and Knowledge

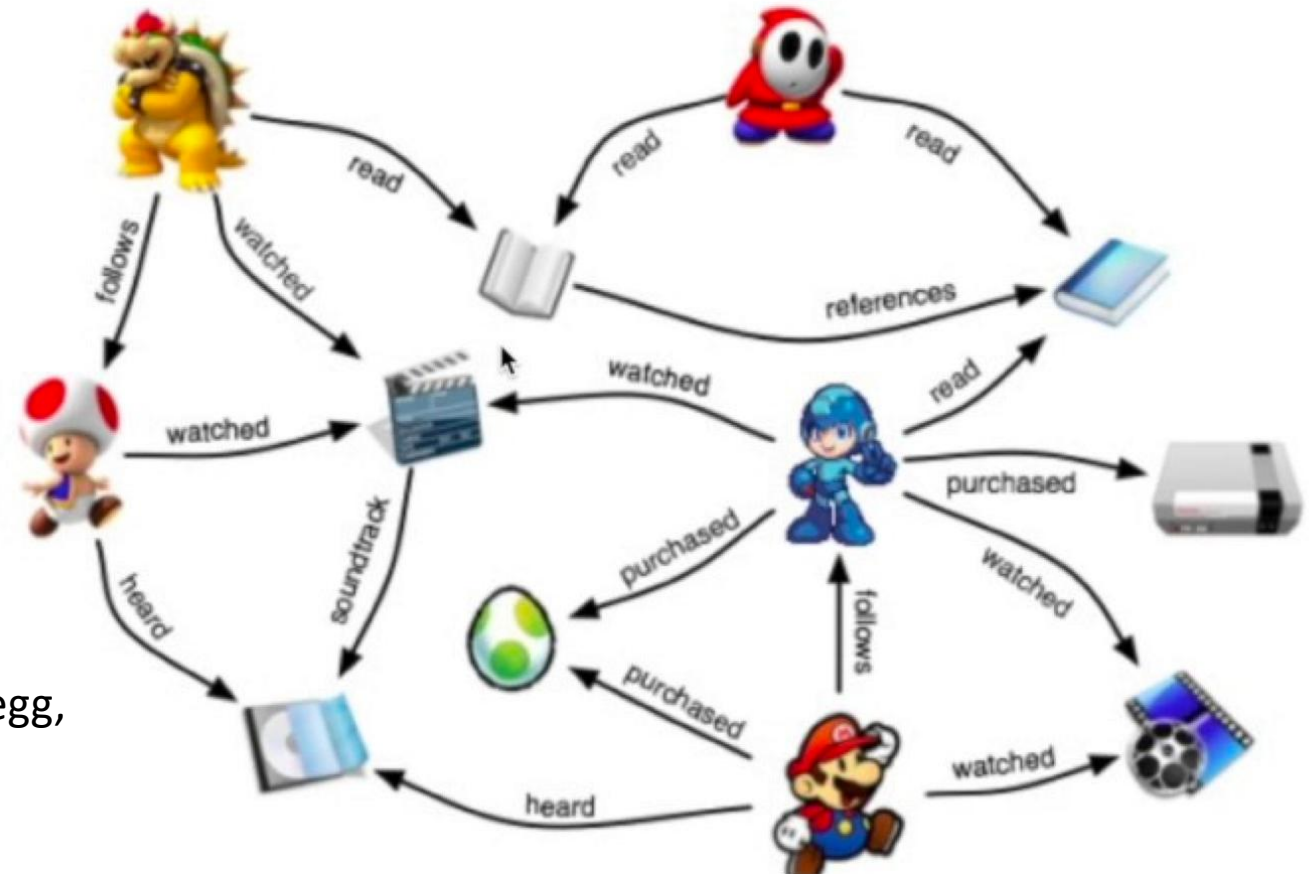
Heterogeneous Graph

Characters: Bowser, Shy Guy, Mega Man, Paper Mario

Media/Objects: Books, movie, CD/soundtrack, a Yoshi egg,

Relationships (Vertices) type of connection:

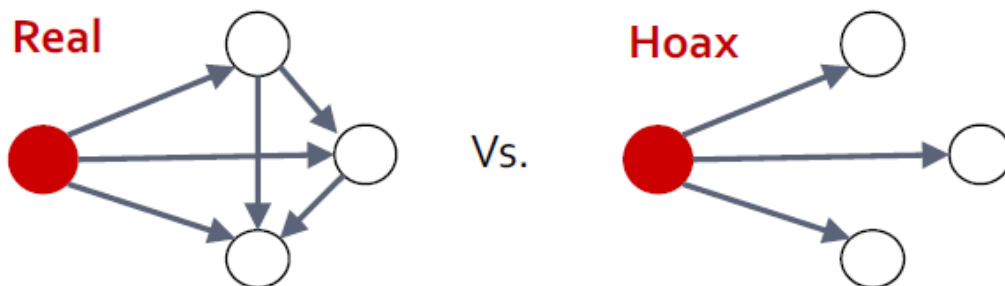
Bowser follows Toad, Bowser read a Book



Application - Misinformation

■ Q: Is a given Wikipedia article a hoax?

- Real articles link more coherently:



Hoax article detection performance:

50%

Random

66%

Human

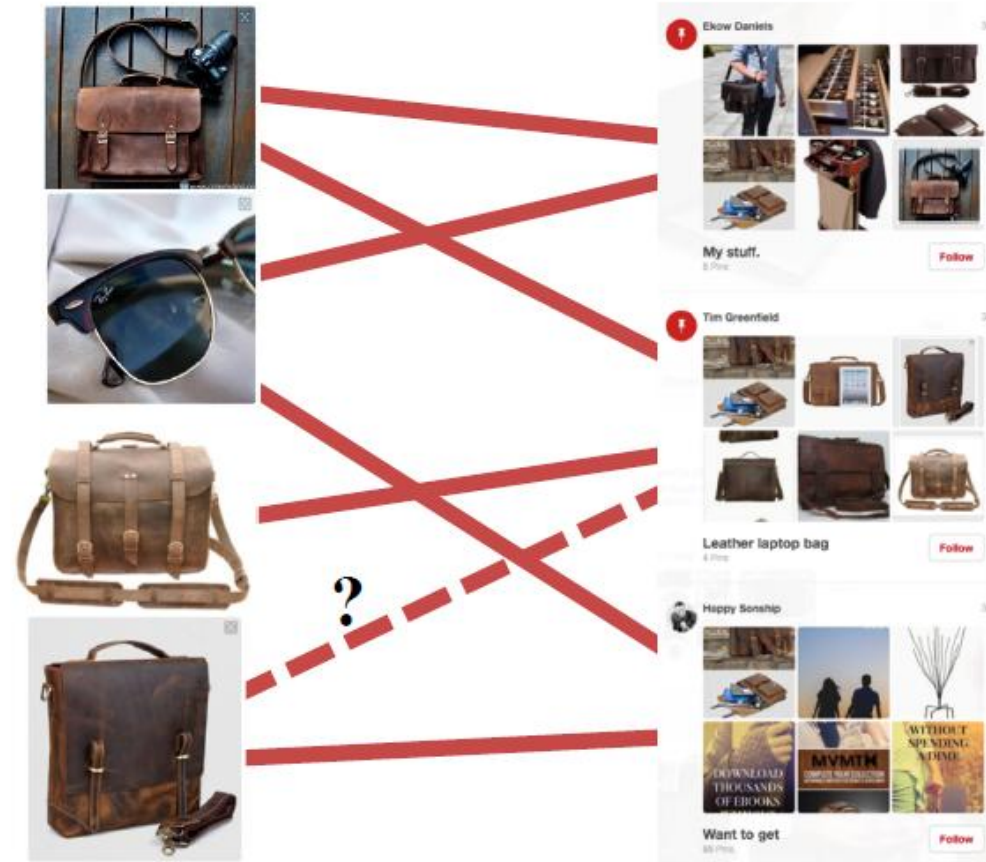
86%

Network

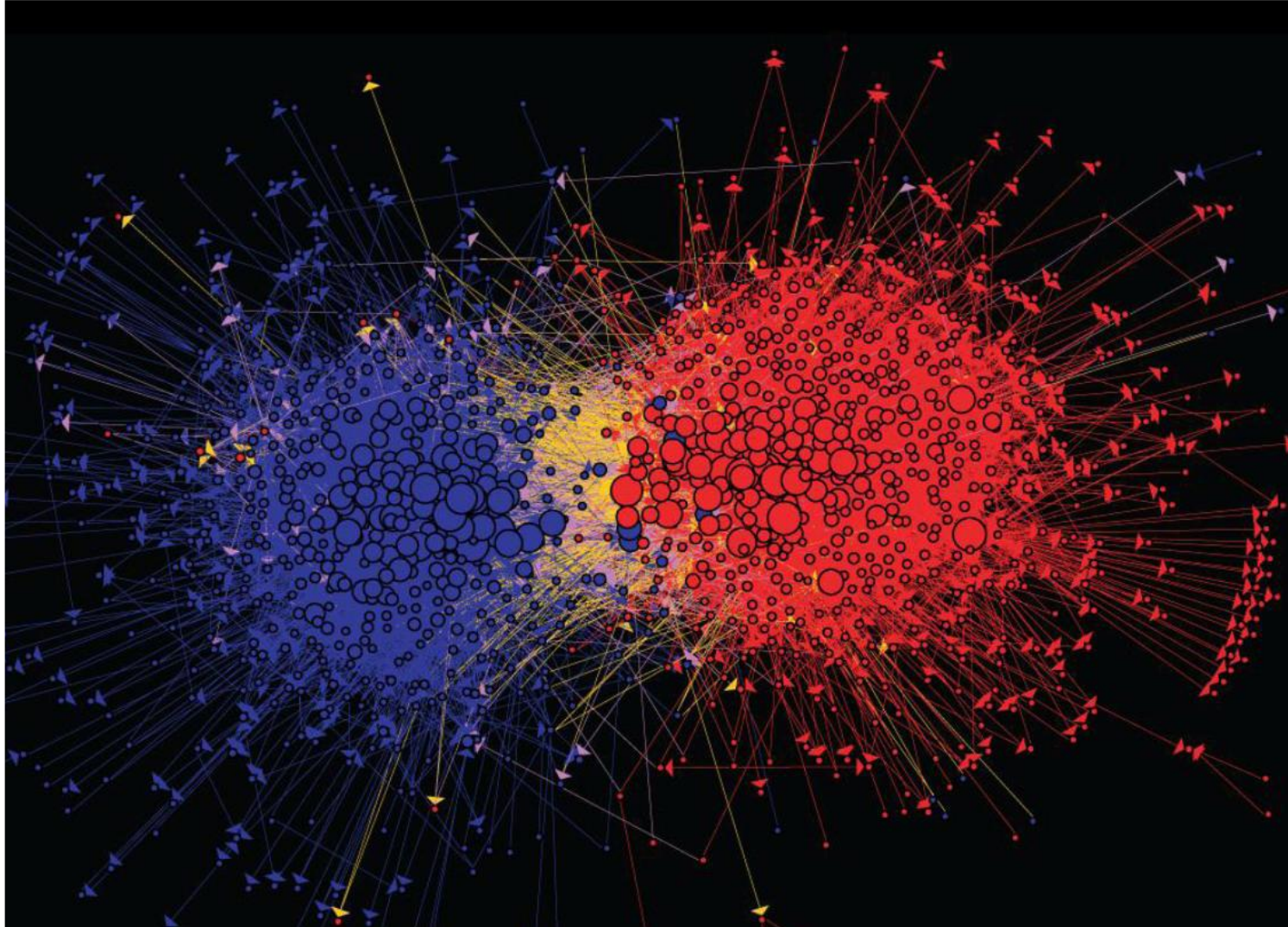


Application - Link Prediction

Content recommendation is **link prediction**

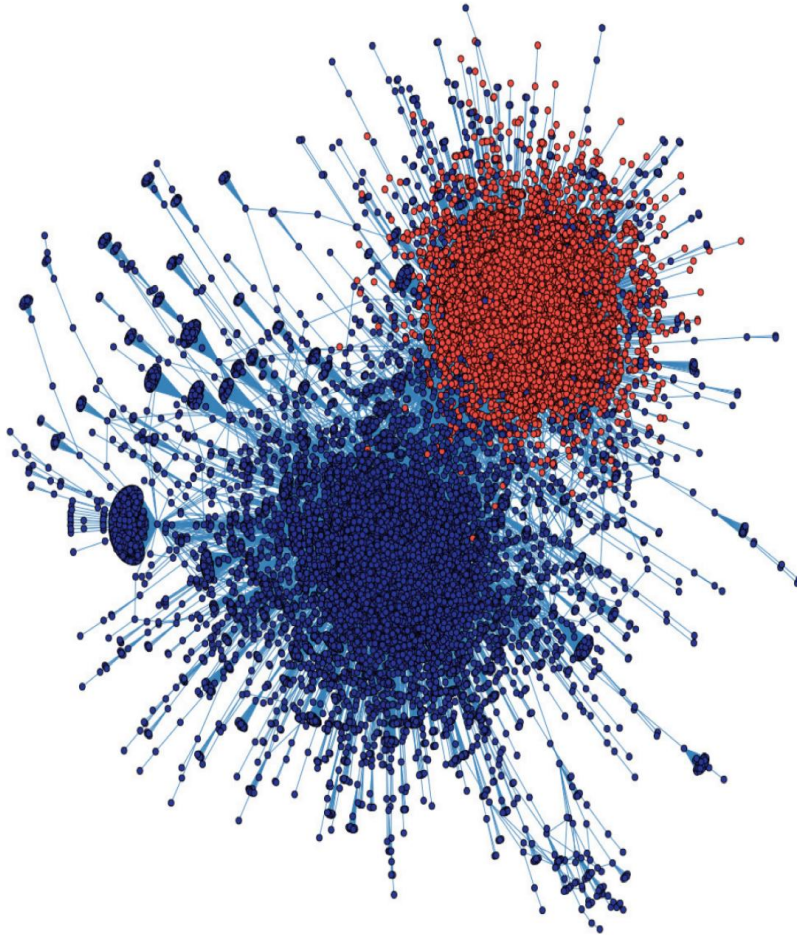


Networks – Online Media

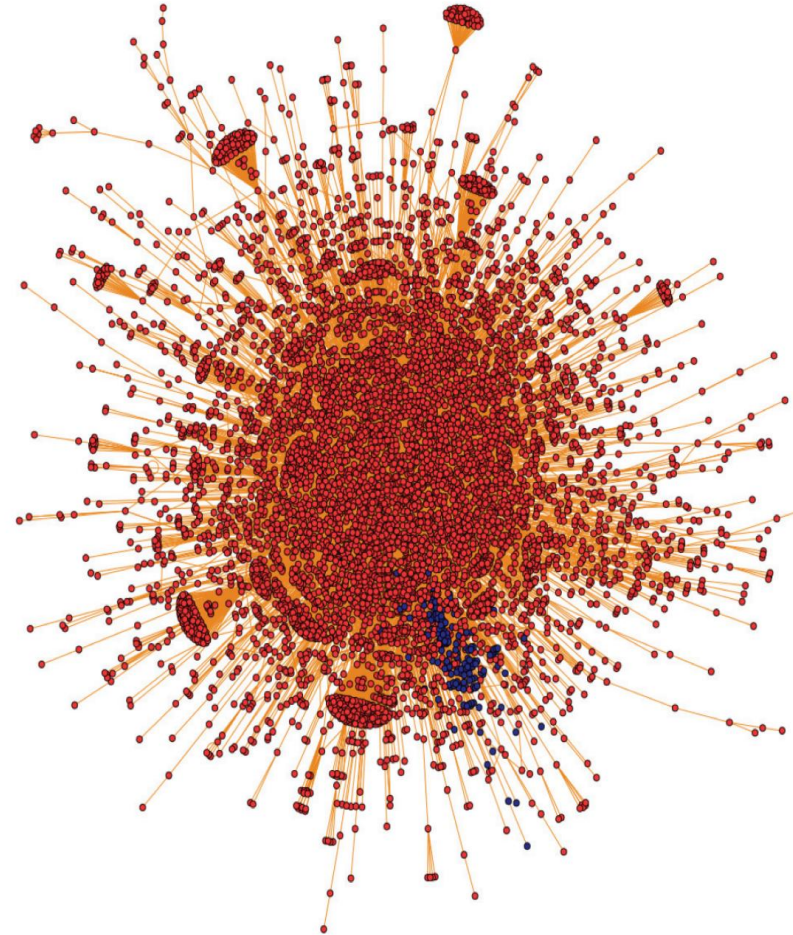


Connections between political blogs

Application - Polarization on Social Media

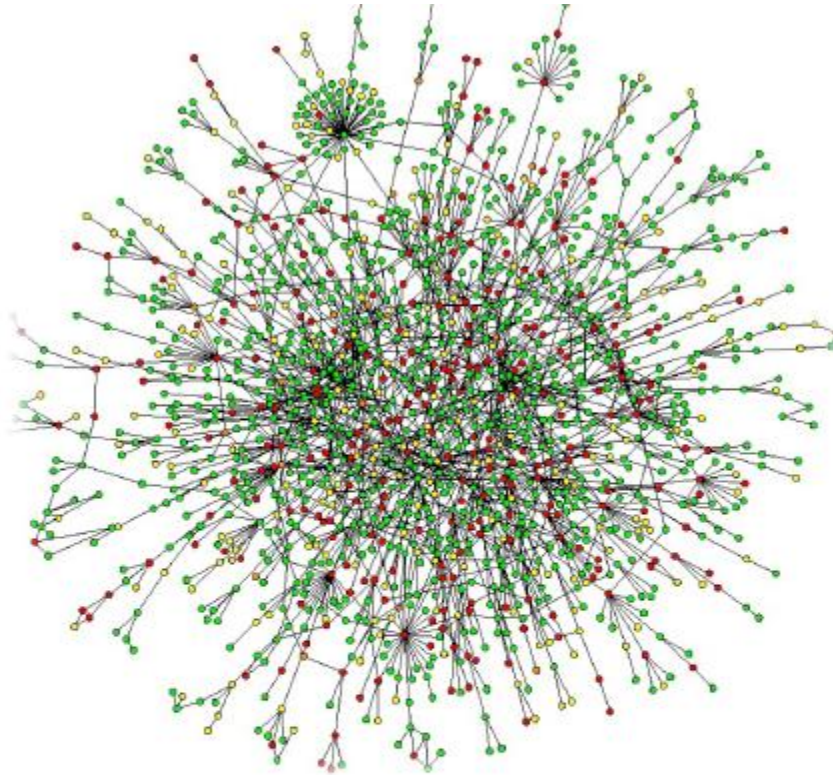


Retweet networks: **Polarized**

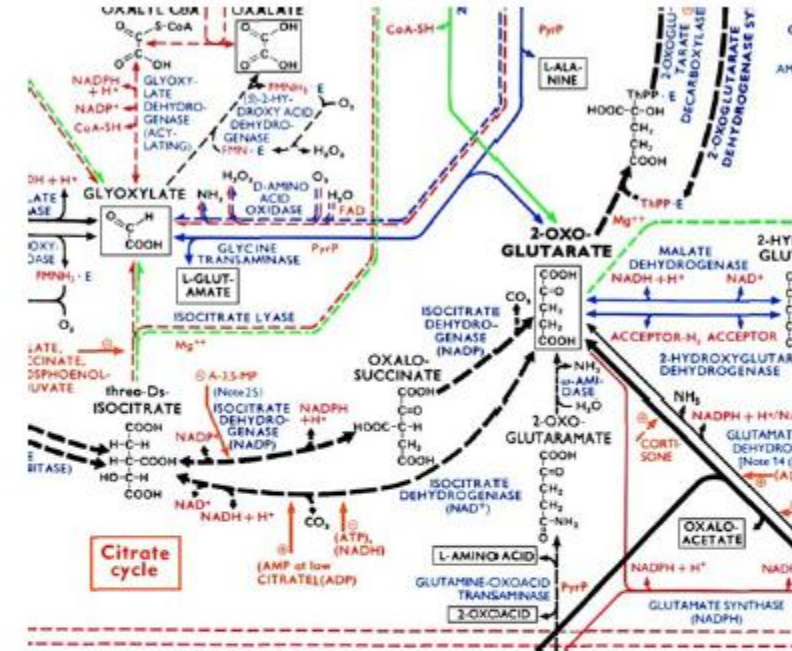


Unpolarized

Networks – Life Science

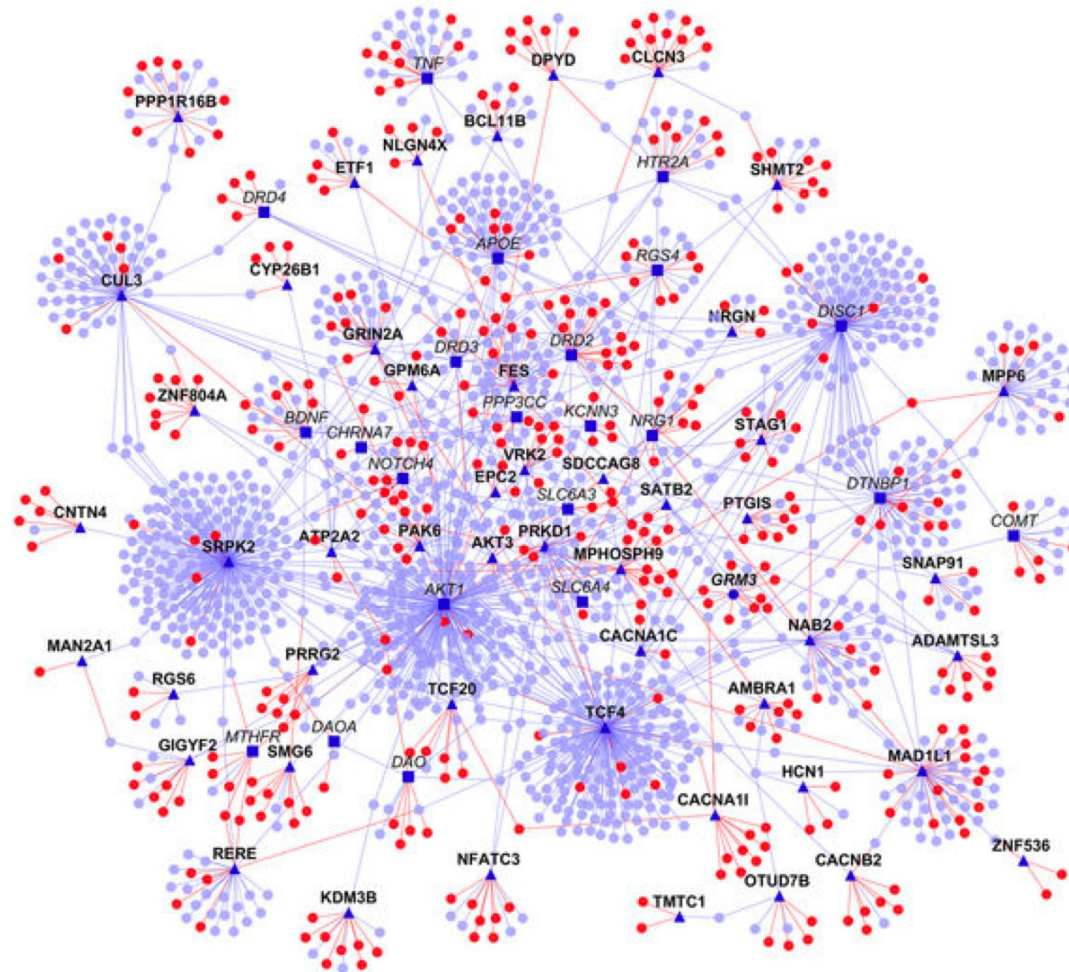


Protein-protein interaction (PPI) networks:
Nodes: Proteins
Edges: 'Physical' interactions



Metabolic networks:
Nodes: Metabolites and enzymes
Edges: Chemical reactions

Application - Biological Function



Classifying the function of proteins in the
Schizophrenia interactome

About the course

Content

- Introduction to networks
- Networks properties
- Various network measures
- Network algorithms
- Network models

About the course

- I assume prior knowledge of python, linear algebra, calculus and basic level of graph theory
- Presentations are in **ENGLISH!** Ask if you don't understand.
- Assignments
 - Implementing algorithms (in Python)
 - Two small assignments (8 pts each)
- Short exam (14 pts) (no Moed B')
- Final project + presentation (50 pts + 20) : **Teams of 3 members**
 - **Each member defends his/her project**

About the Project

- Final Project : **Teams of 3 members**
- Start thinking about the project **NOW**
- Tools:
 - NetworkX
 - iGraph
 -

Introduction

- Introduce the basic mathematical tools to understand the fundamentals of **complex networks**
- Provide the skills that are needed to perform basic analysis of such networks
- Learn algorithms to analyze and understand the properties of networks

What are these networks

- Many real-world **systems** can be viewed as a collection of **nodes** that are **linked** to each other.
- To name but a few:
 - Traffic infrastructure: roads, railways, shipping, airlines
 - Social communities: family ties, online communities
 - Communication networks: Internet, telecommunication
 - Biological and chemical systems

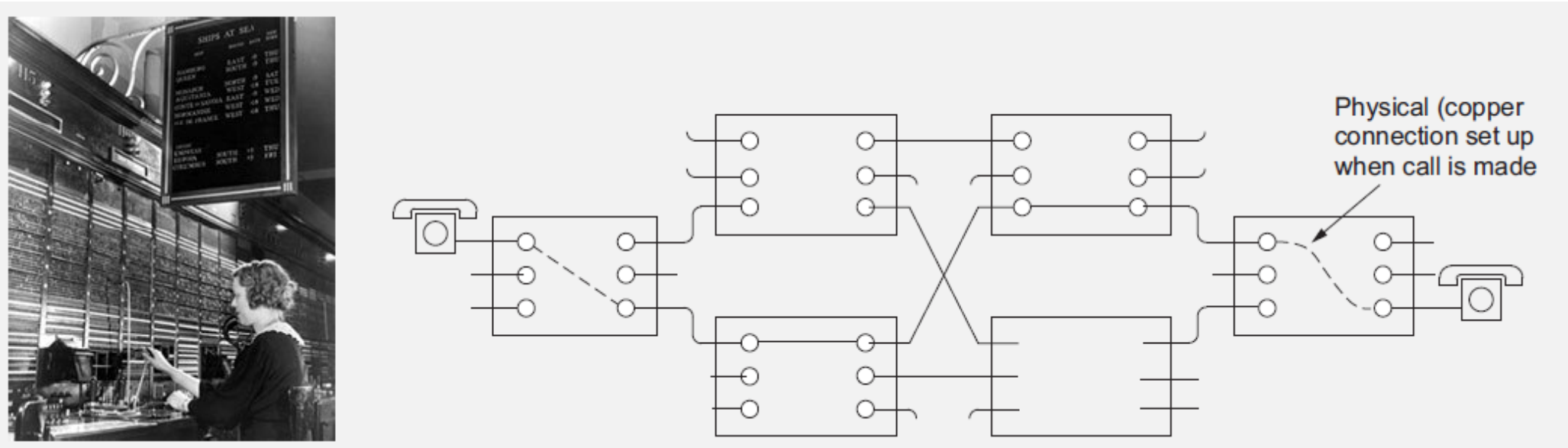
Data -> graph ?

Connected world

- In the **very old days**: carriers of messages (pigeons, horses, etc.)
- Also in the **old days**: fire beacons, mirrors, drums, flags.
- By the 1850s, communication was carried over more than 30,000 kms of electrical telegraph.
- Since the late 1900s (20'th cent.) : communication networks

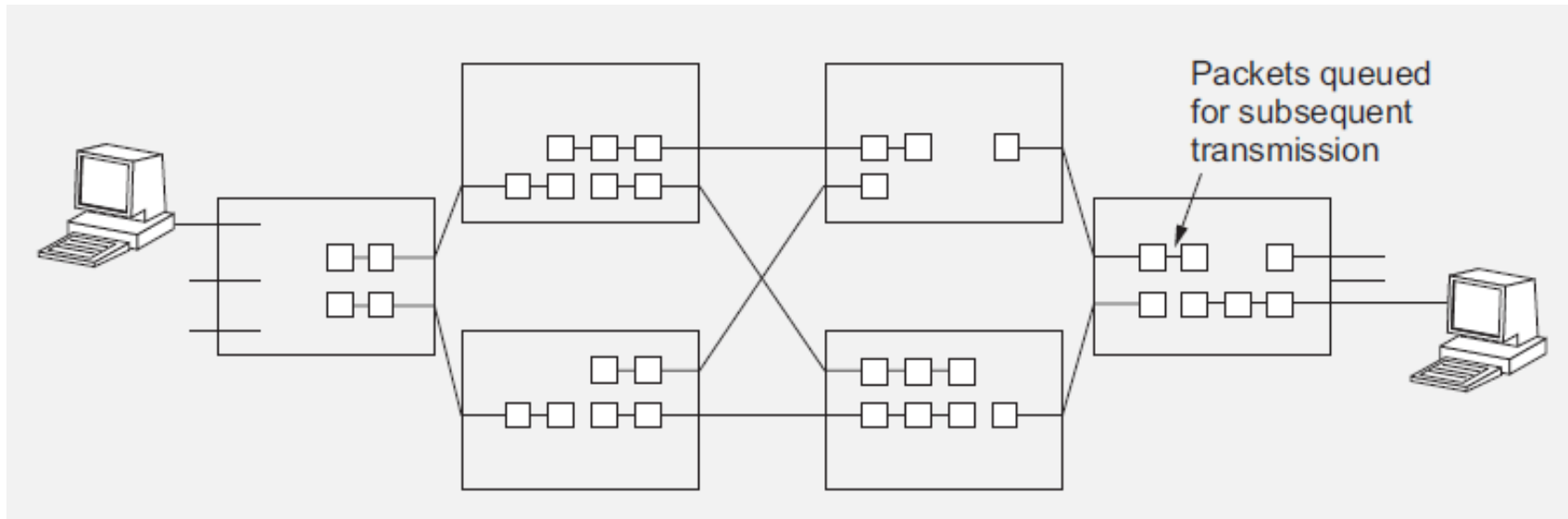
Connected world

- In traditional telecommunications networks, to hold a conversation, it was necessary to make a **physical connection** between the two parties → **circuit-switched network**.



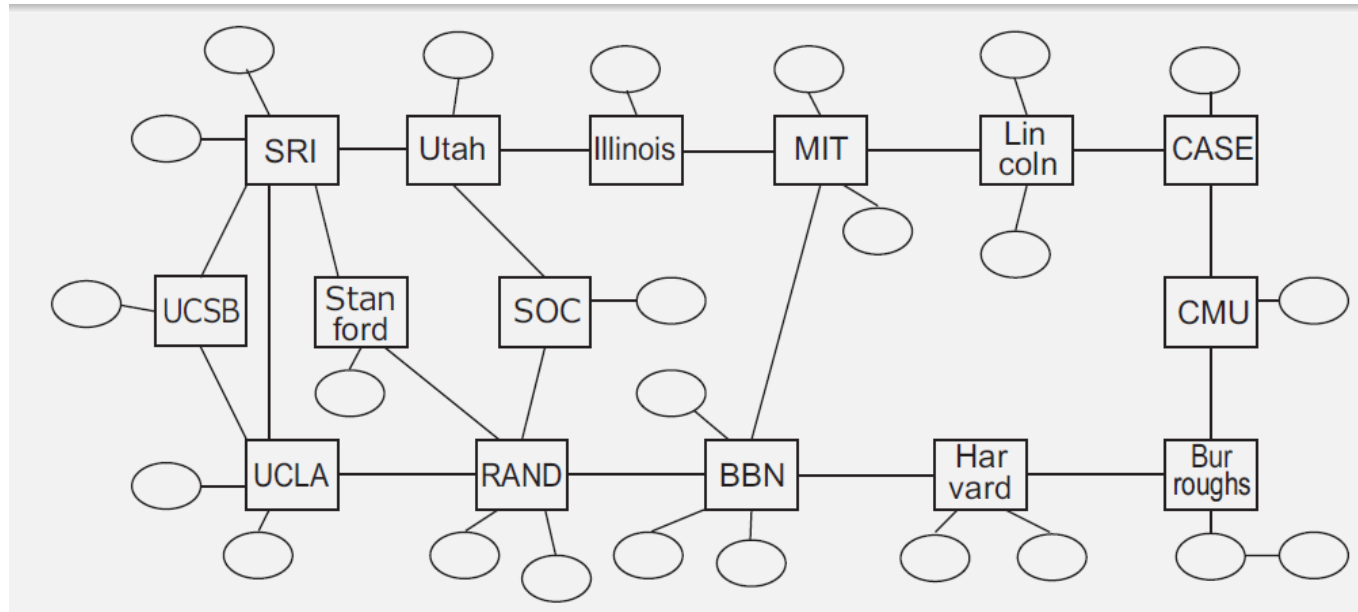
Connected world – Modern networks

- In modern **telephony** networks, everything is packetized:
 - Data (including **samples** from continuous media) is put into a **packet**.
 - Packets are extended with destination address and are independently **routed**.



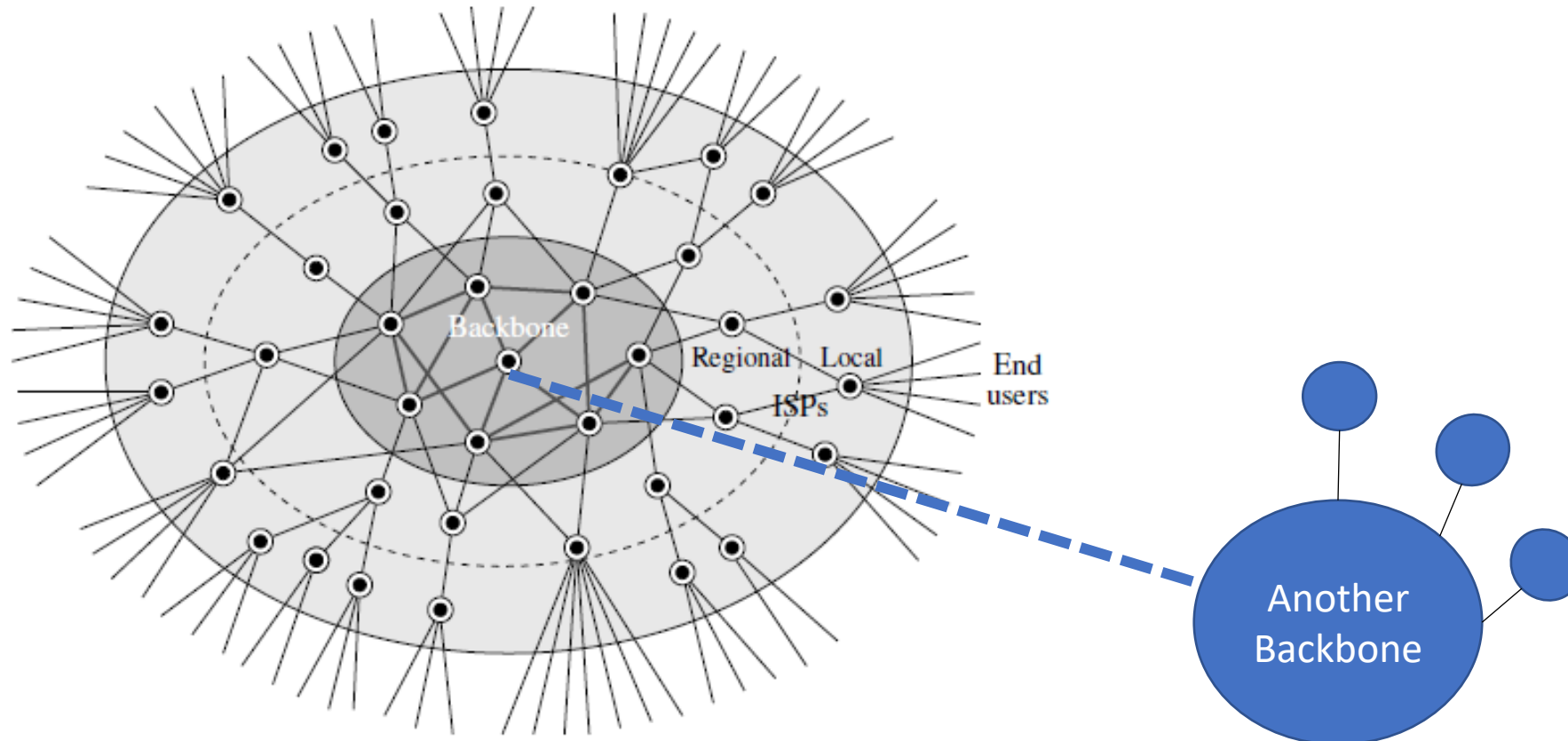
The Internet

- The **Internet** connect many computers through switches that automatically discover and maintain routes



The Internet – schematic view

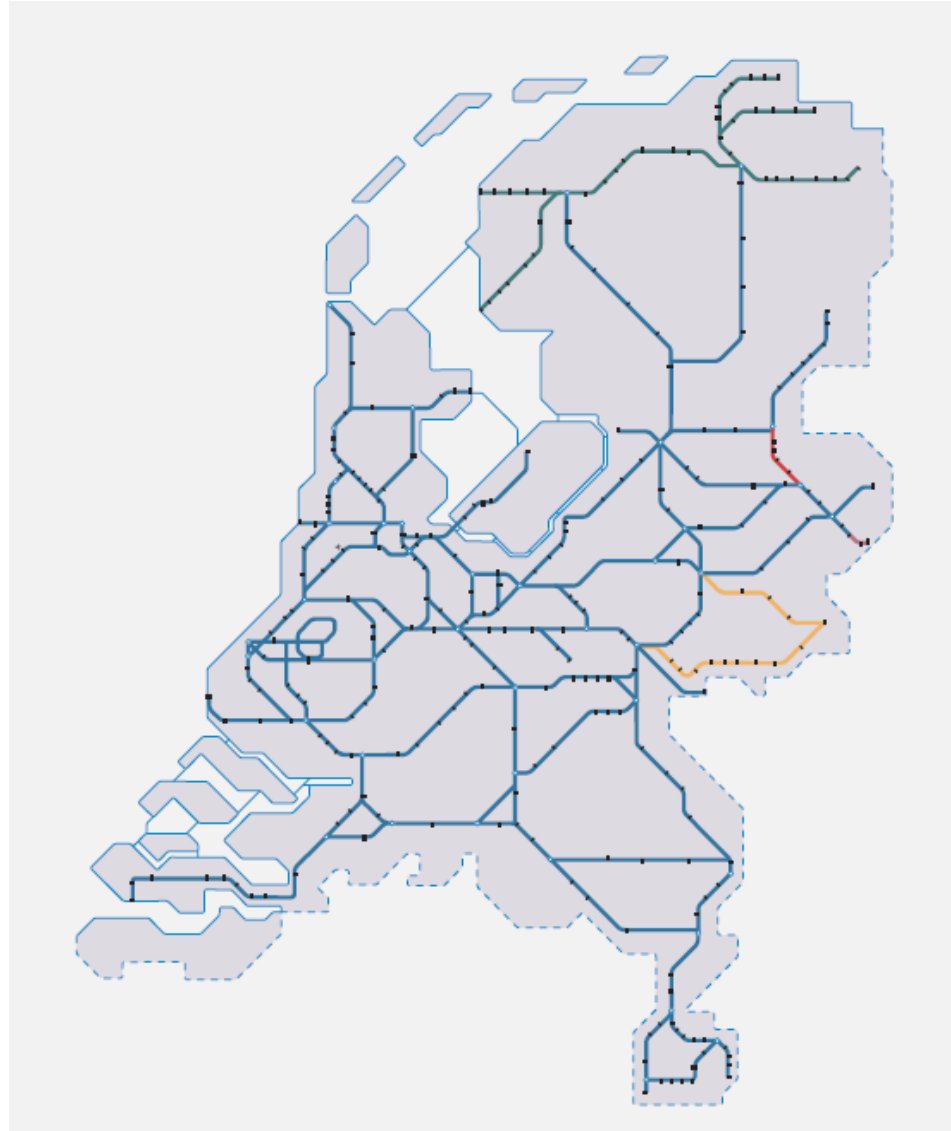
- The nodes and edges of the Internet can be divided to **the backbone** (high-bandwidth long-distance connections), **the ISPs** who connect to the backbone and the **end users** who connect to the ISPs



Network examples

Railway system

Why this is an example?



Network examples

Airline flight routes

United



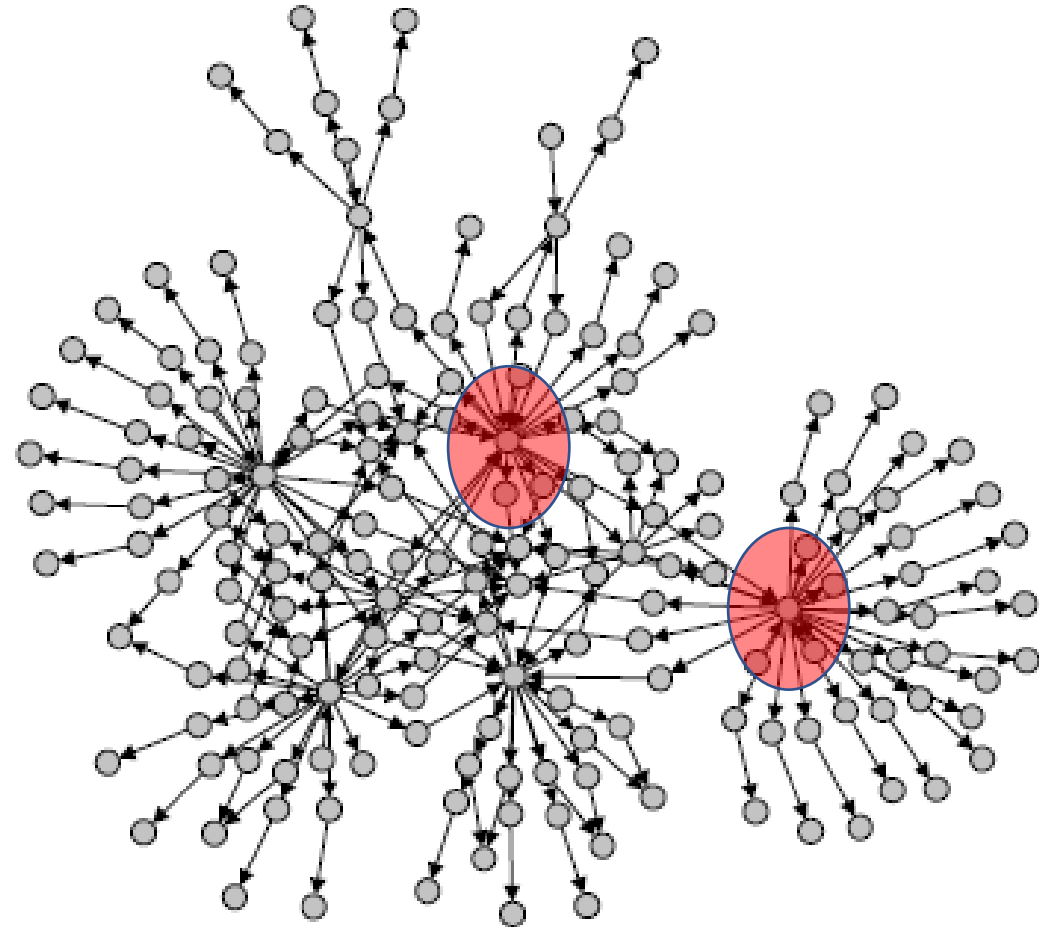
Continental



Question: **What main differences can be observed?**

WWW network

- Example of connections between web pages on a single website
 - How do we find important nodes (hub)
 - How do we measure their influence

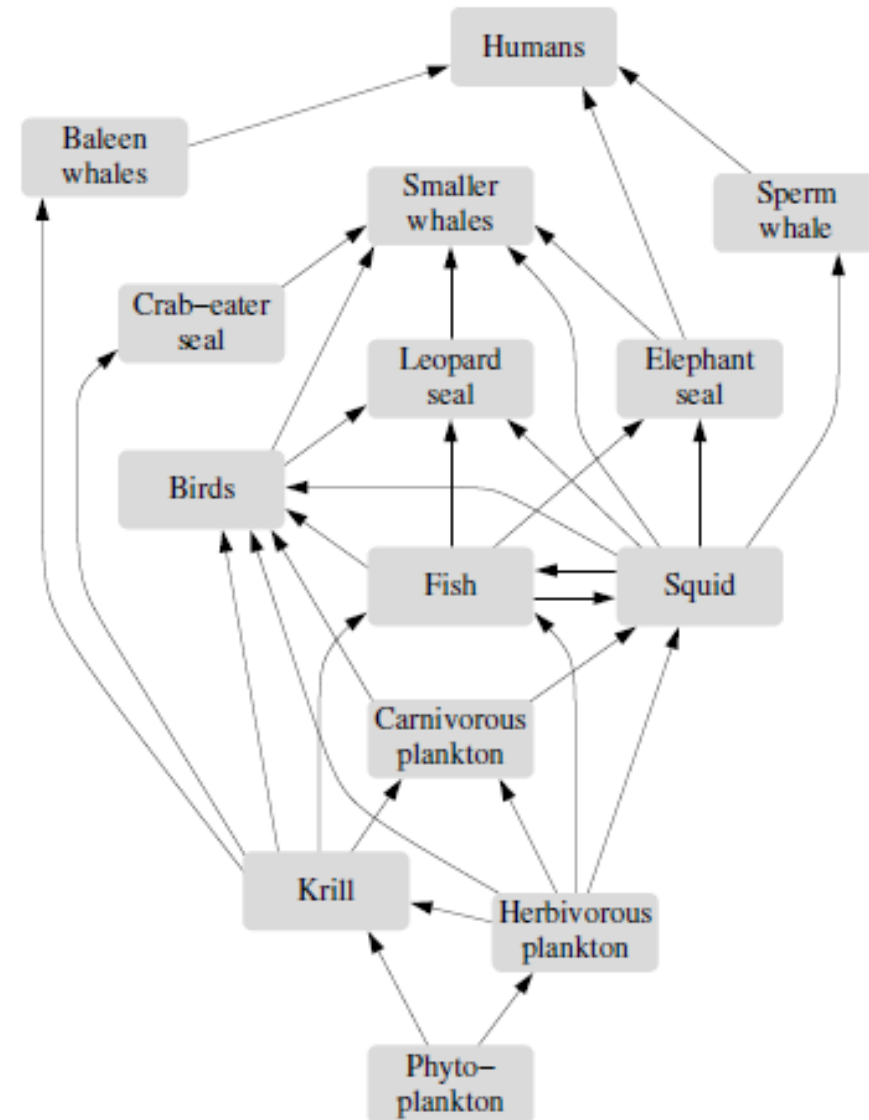


- Brain interactions in C-elegance (out of total of 302 neurons)



Food-chain (ecological) network

- Predator – Prey interactions in Antarctica

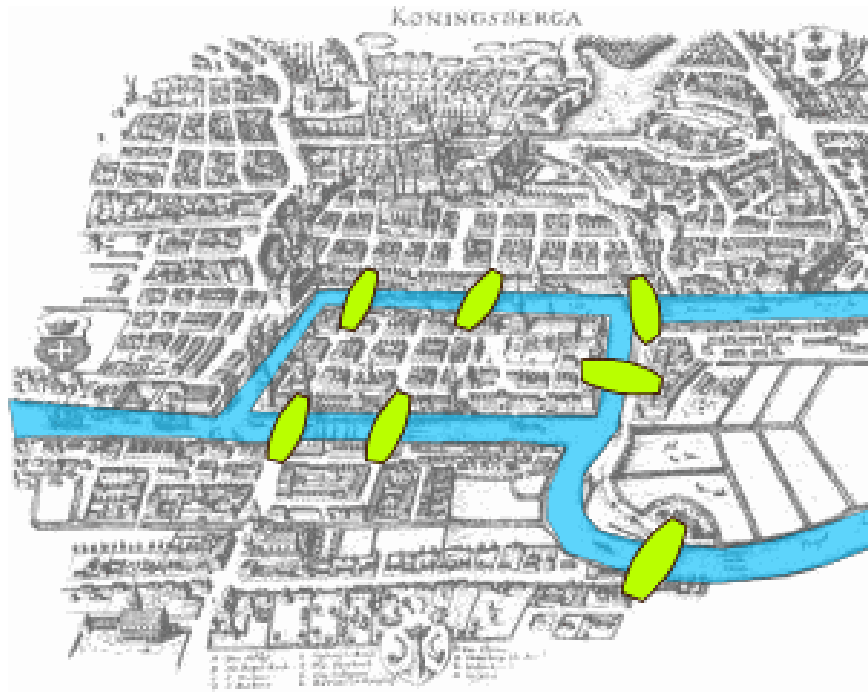


Nodes and edges examples in networks

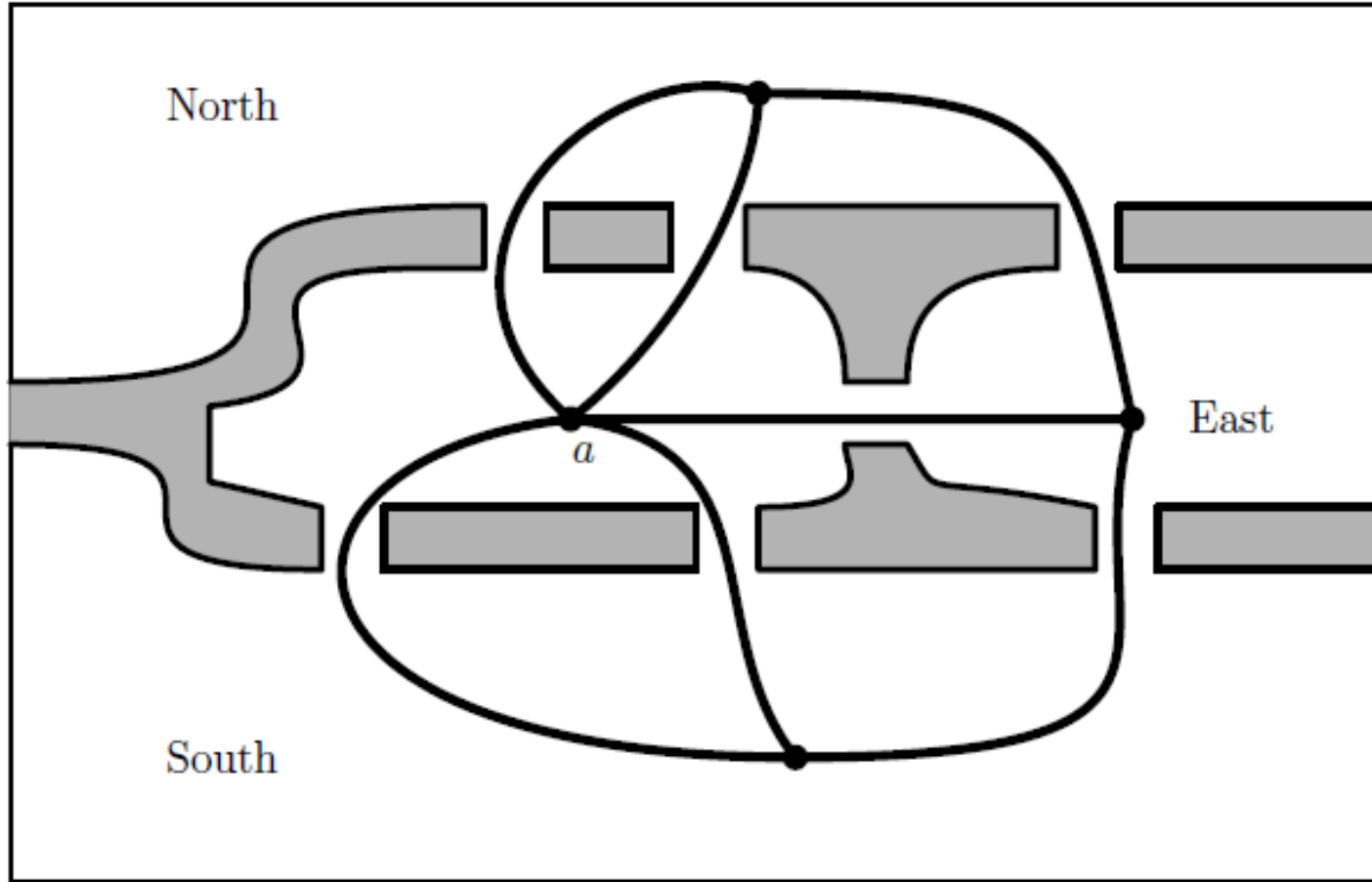
Network	Node	Edge
Internet	Computer or router	Cable or wireless data connection
World Wide Web	Web page	Hyperlink
Citation network	Article, patent, or legal case	Citation
Power grid	Generating station or substation	Transmission line
Friendship network	Person	Friendship
Metabolic network	Metabolite	Metabolic reaction
Neural network	Neuron	Synapse
Food web	Species	Predation

The Königsberg bridge problem

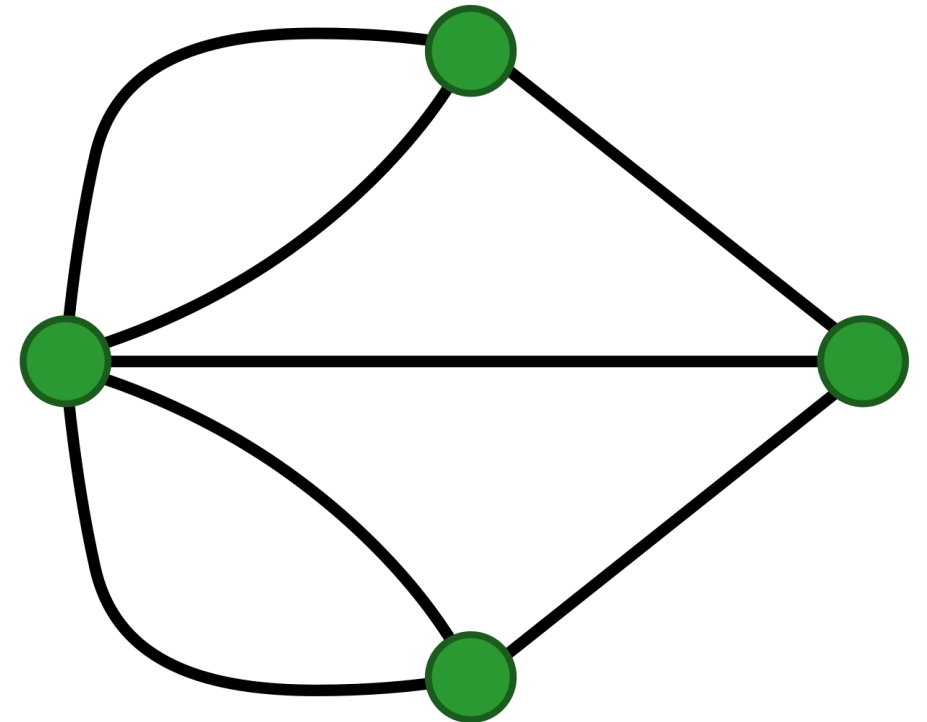
- Graph theory began in 1736 when Leonhard Euler solved the well known Königsberg bridge problem (7 bridges of Königsberg)
- The problem was to devise a walk through the city that would cross each of those bridges once and only once.



The problem as a graph



Bridge => Edge



Preliminaries

Networks or Graphs

- **Network** often refers to real systems
 - Web, Social network, Metabolic network

Language: Network, node, link

- **Graph** is a mathematical representation of a network
 - Web graph, Social graph (a Facebook term)

Language: Graph, vertex, edge

In most cases we will use the two terms interchangeably

Preliminary Foundations

Graph Definitions

A **graph** G is a tuple (V, E) of **vertices** V and a collection of **edges** E . Each edge $e \in E$ is said to connect two vertices $u, v \in V$, and is denoted as $e = \langle u, v \rangle$.

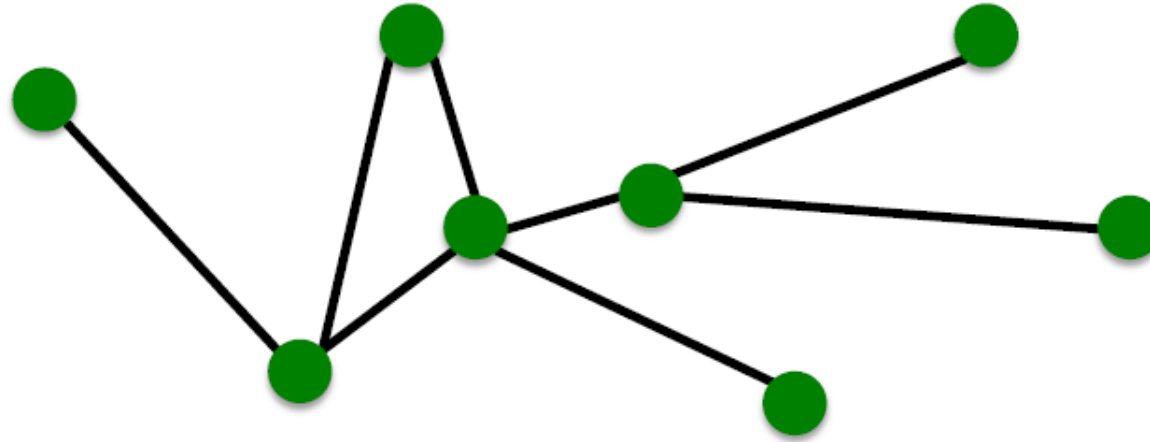
Notations: $V(G)$, $E(G)$.

The **complement** $\bar{G}$ of a graph G , has the same vertex set as G , but $e \in E(\bar{G})$ *if and only if* $e \notin E(G)$.

For any graph G and vertex $v \in V(G)$, the **neighbor set** $N(v)$ of v is the set of vertices (other than v) adjacent to v :

$$N(v) = \{w \in V(G) \mid v \neq w, \langle v, w \rangle \in E(G)\}$$

Graph/Network Components



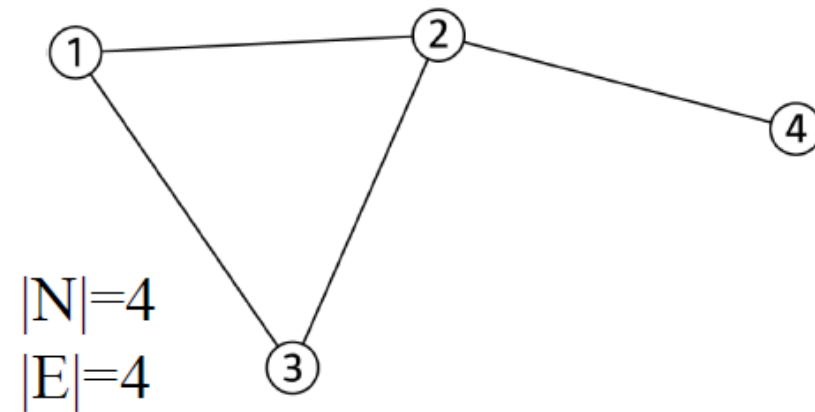
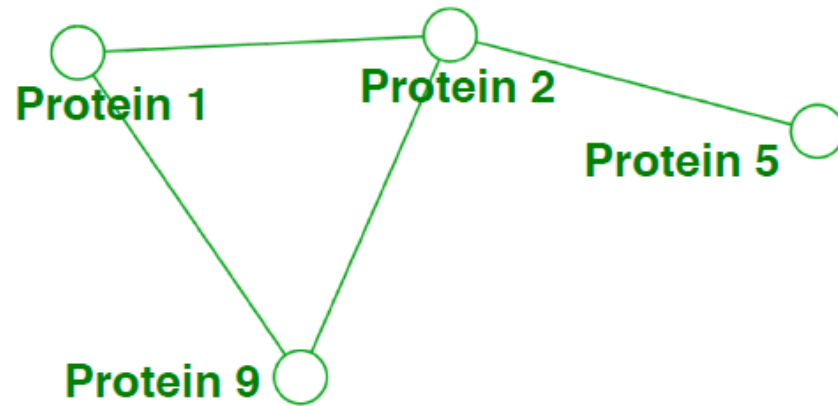
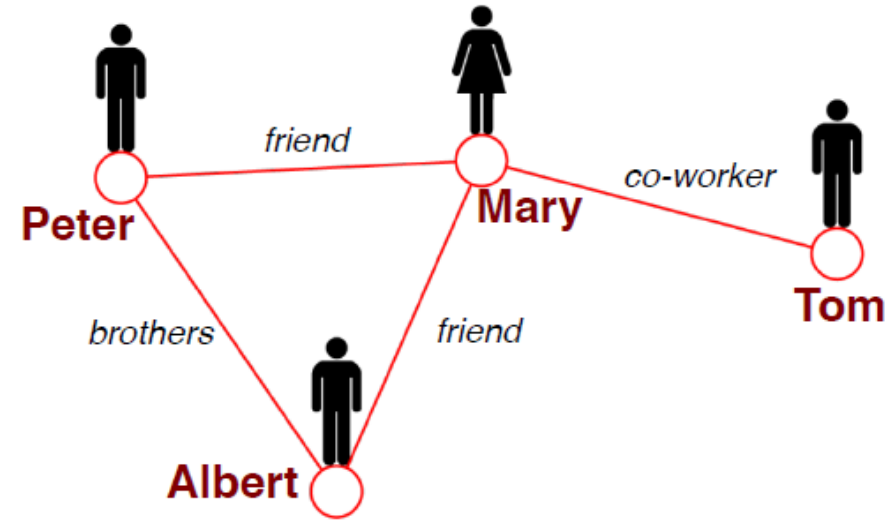
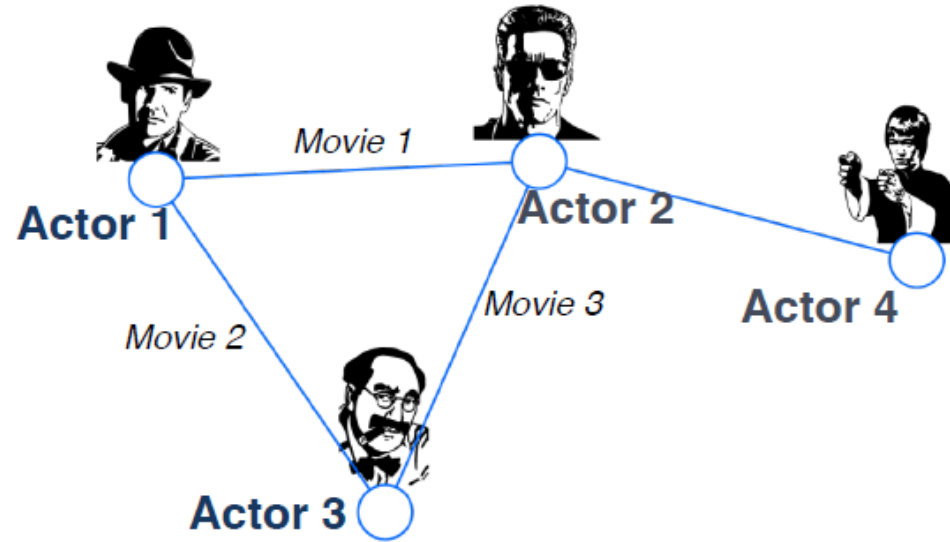
- **Objects:** nodes, vertices
- **Interactions:** links, edges
- **System:** network, graph

N

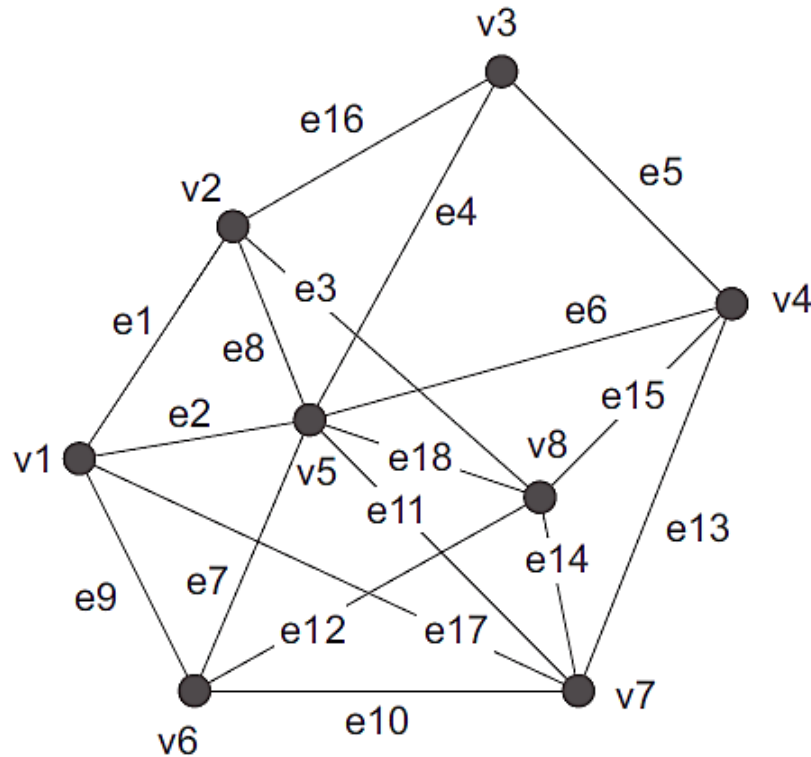
E

$G(N,E)$

Common Language



Graph example



$$V(G) = \{v_1, \dots, v_8\}$$

$$E(G) = \{e_1, \dots, e_{18}\}$$

$$e_1 = \langle v_1, v_2 \rangle \quad e_{10} = \langle v_6, v_7 \rangle$$

$$e_2 = \langle v_1, v_5 \rangle \quad e_{11} = \langle v_5, v_7 \rangle$$

$$e_3 = \langle v_2, v_8 \rangle \quad e_{12} = \langle v_6, v_8 \rangle$$

$$e_4 = \langle v_3, v_5 \rangle \quad e_{13} = \langle v_4, v_7 \rangle$$

$$e_5 = \langle v_3, v_4 \rangle \quad e_{14} = \langle v_7, v_8 \rangle$$

$$e_6 = \langle v_4, v_5 \rangle \quad e_{15} = \langle v_4, v_8 \rangle$$

$$e_7 = \langle v_5, v_6 \rangle \quad e_{16} = \langle v_2, v_3 \rangle$$

$$e_8 = \langle v_2, v_5 \rangle \quad e_{17} = \langle v_1, v_7 \rangle$$

$$e_9 = \langle v_1, v_6 \rangle \quad e_{18} = \langle v_5, v_8 \rangle$$

- What is the neighbor set of v_1 - $N(v_1)$?

Nodes Degree

- The number of edges incident with a node/vertex v is called the **Degree** of v , denoted as $\delta(v)$ (sometimes $d(v)$).

Loops, i.e., edges joining a vertex with itself, are counted twice.

- Theorem

For all graphs G , $\sum_{v \in V(G)} \delta(v)$ is $2 \cdot |E(G)|$.

i.e. the sum of nodes degree is twice the number of edges Why?

Subgraphs

- Definitions

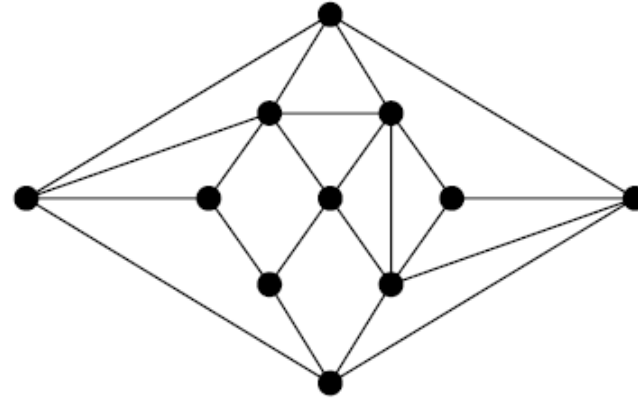
H is a **subgraph** of G if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$ such that for all $e \in E(H)$ with $e = \langle u, v \rangle : u, v \in V(H)$.

The **subgraph induced by** $V^* \subseteq V(G)$ has vertex set V^* and edge set $\{\langle v, w \rangle \in E(G) \mid v, w \in V^*\}$. Denoted as $H = G[V^*]$. The **subgraph induced by** $E^* \subseteq E(G)$ has vertex set $V(G)$ and edge set E^* . Denoted as $H = G[E^*]$.

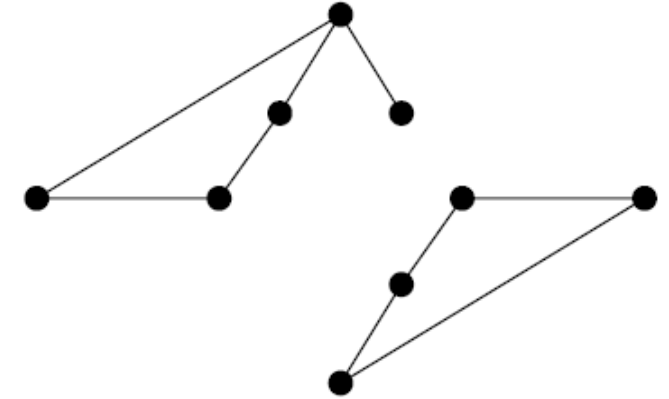
Subgraphs examples

- An induced subgraph includes **all the edges** that have **both endpoints** in the inducing set $\{S\}$,

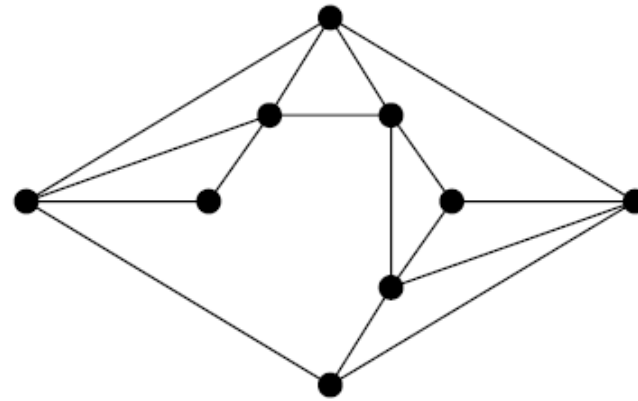
whereas an ordinary subgraph may miss some.



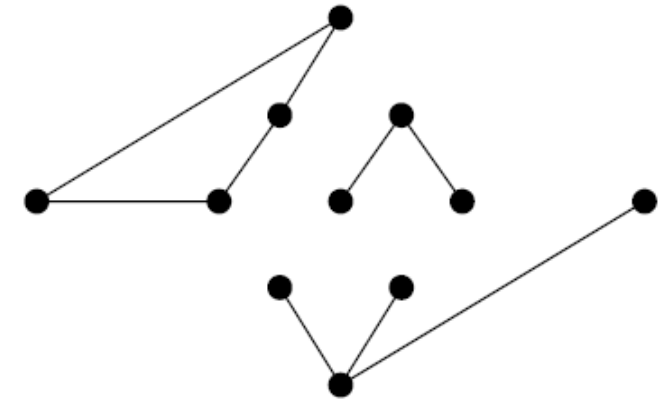
a graph



a subgraph



an induced subgraph



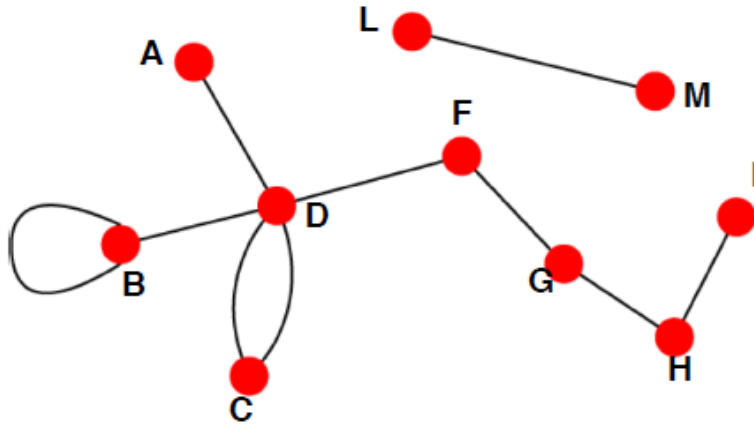
a spanning subgraph

Networks Representation

Directed / Undirected Graphs

Undirected

- **Links:** undirected (symmetrical, reciprocal)

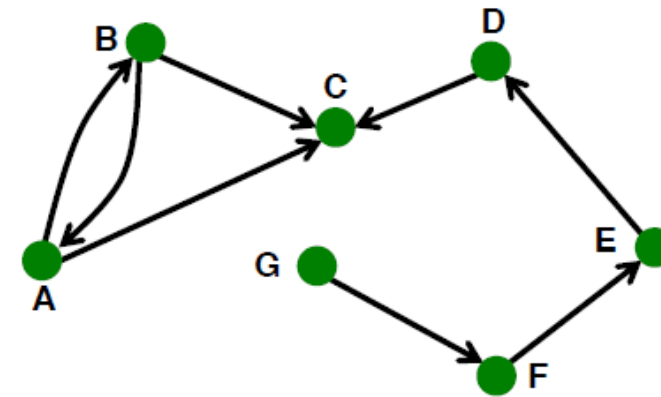


- **Examples:**

- Collaborations
- Friendship on Facebook

Directed

- **Links:** directed (arcs)

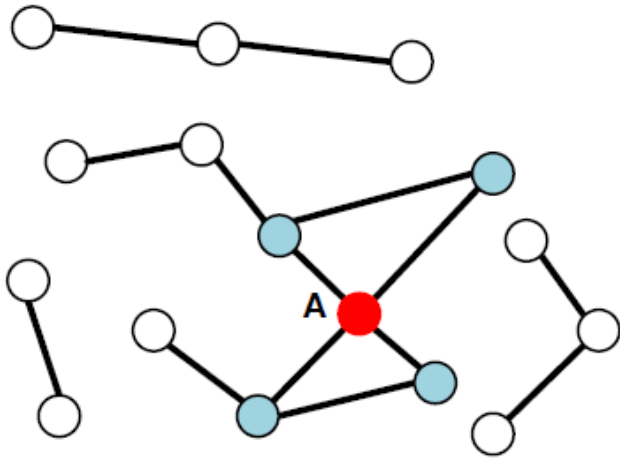


- **Examples:**

- Phone calls
- Following on Twitter

Node Degrees

Undirected



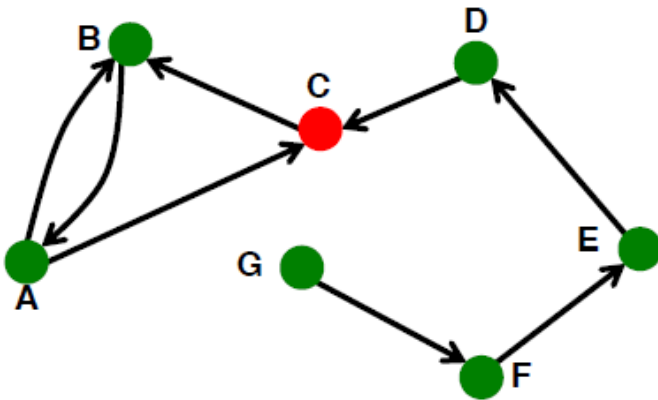
Node degree, k_i : the number of edges adjacent to node i

$$k_A = 4$$

Node degree is also denoted by d_i

Avg. degree: $\bar{k} = \langle k \rangle = \frac{1}{N} \sum_{i=1}^N k_i = \frac{2E}{N}$

Directed



Source: Node with $k^{in} = 0$

Sink: Node with $k^{out} = 0$

In directed networks we define an **in-degree** and **out-degree**.

The (total) degree of a node is the sum of in- and out-degrees.

$$k_C^{in} = 2 \quad k_C^{out} = 1 \quad k_C = 3$$

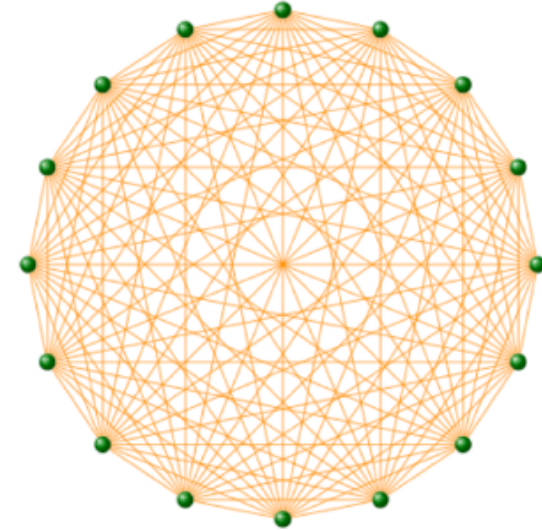
$$\bar{k} = \frac{E}{N}$$

$$\overline{k^{in}} = \overline{k^{out}}$$

Complete Graph

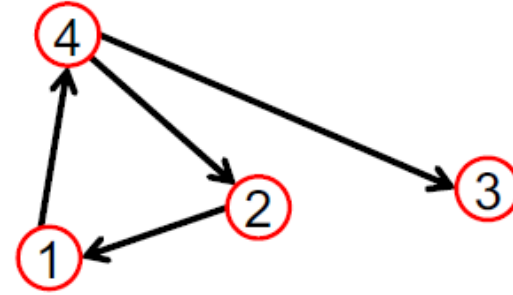
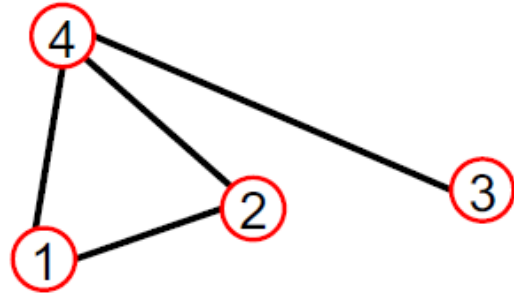
The **maximum number of edges** in an undirected graph on N nodes is

$$E_{\max} = \binom{N}{2} = \frac{N(N-1)}{2}$$



An undirected graph with the number of edges $E = E_{\max}$ is called a **complete graph**, and its average degree is $N-1$

Adjacency Matrix



$A_{ij} = 1$ if there is a link from node i to node j

$A_{ij} = 0$ otherwise

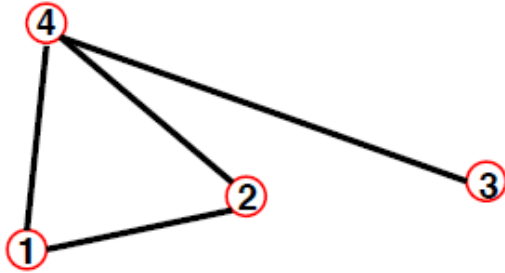
$$A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

$$A = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

Note that for a directed graph (right) the matrix is not symmetric

Adjacency Matrix

Undirected



$$A_{ij} = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

$$A_{ij} = A_{ji}$$

$$A_{ii} = 0$$

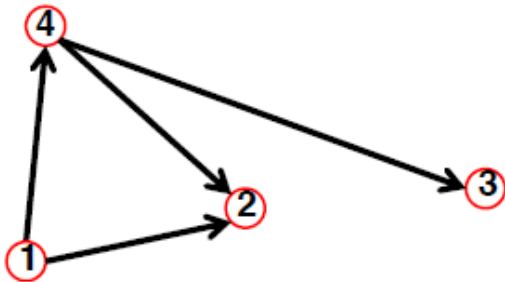
$$k_i = \sum_{j=1}^N A_{ij}$$

$$k_j = \sum_{i=1}^N A_{ij}$$

$$L = \frac{1}{2} \sum_{i=1}^N k_i = \frac{1}{2} \sum_{ij} A_{ij}$$

L = No. of edges

Directed



$$A = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

$$A_{ij} \neq A_{ji}$$

$$A_{ii} = 0$$

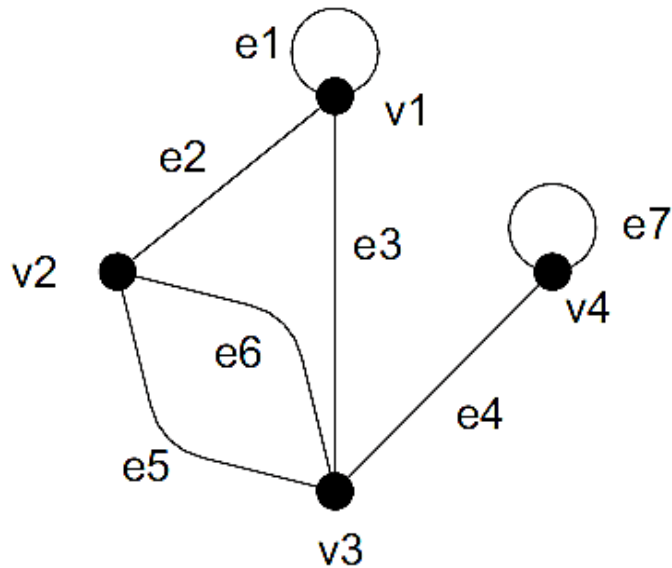
$$k_i^{out} = \sum_{j=1}^N A_{ij}$$

$$k_j^{in} = \sum_{i=1}^N A_{ij}$$

$$L = \sum_{i=1}^N k_i^{in} = \sum_{j=1}^N k_j^{out} = \sum_{i,j} A_{ij}$$

Incidence Matrix

Alternative description

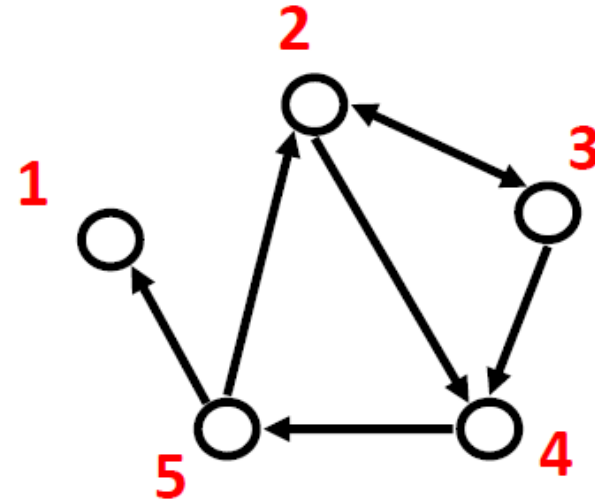


	e_1	e_2	e_3	e_4	e_5	e_6	e_7
V_1	2	1	1	0	0	0	0
V_2	0	1	0	0	1	1	0
V_3	0	0	1	1	1	1	0
V_4	0	0	0	1	0	0	2

Graph Representation – Edge List

- **Represent graph as a set of edges:**

- (2, 3)
- (2, 4)
- (3, 2)
- (3, 4)
- (4, 5)
- (5, 2)
- (5, 1)



Graph Representation – Adjacency List

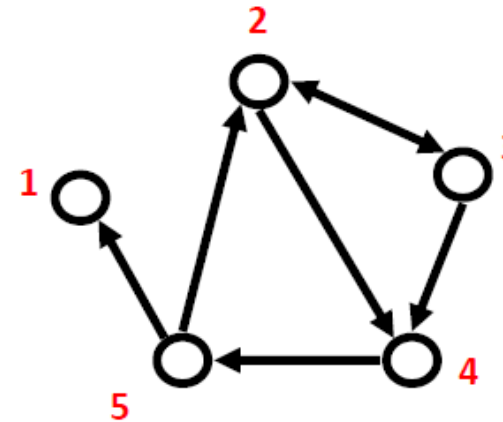
Adjacency list:

- Easier to work with if network is

- Large
- Sparse

- Allows us to quickly retrieve all neighbors of a given node

- 1:
- 2: 3, 4
- 3: 2, 4
- 4: 5
- 5: 1, 2



Most real-world networks are **sparse**

$$E \ll E_{\max} \text{ (or } k \ll N-1)$$

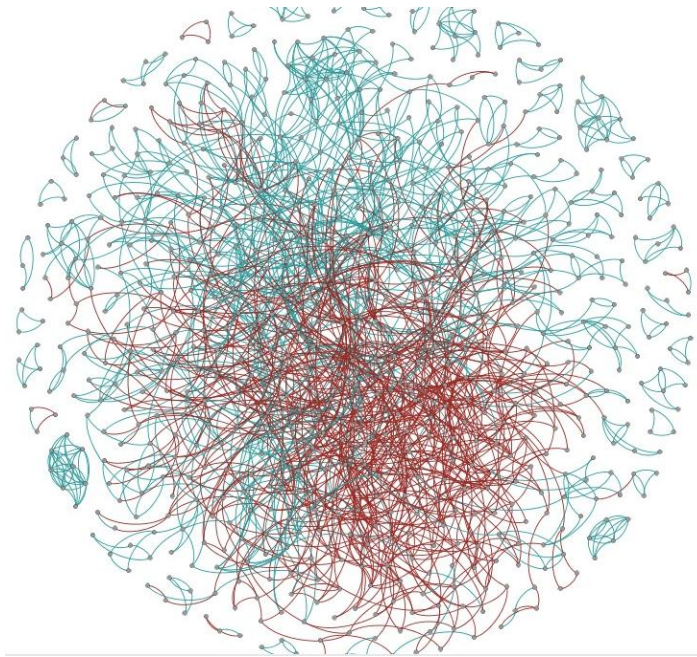
Consequence: Adjacency matrix is filled with zeros!

(Density of the matrix (E/N^2): $WWW \approx 1.5 \times 10^{-5}$)

Edge Attributes

Possible options:

- Weight (e.g. frequency of communication)
- Ranking (best friend, second best friend...)
- Type (friend, relative, co-worker)
- Sign: Friend vs. Foe, Trust vs. Distrust
- Properties depending on the structure of the rest of the graph:
number of common friends



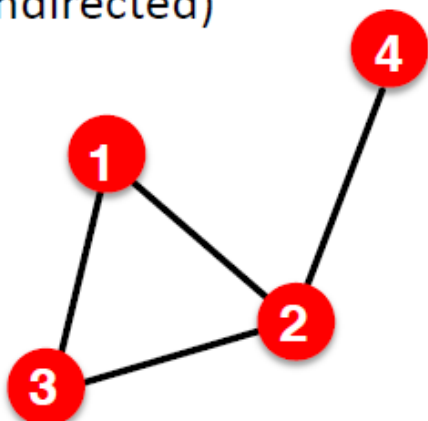
sample of positive & negative ratings from Epinions network

One person trusting/distrusting another

Weights

■ Unweighted

(undirected)



$$A_{ij} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

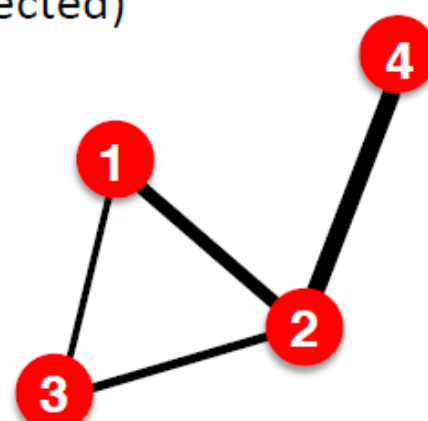
$$A_{ii} = 0$$

$$A_{ij} = A_{ji}$$

$$E = \frac{1}{2} \sum_{i,j=1}^N A_{ij} \quad \bar{k} = \frac{2E}{N}$$

■ Weighted

(undirected)



$$A_{ij} = \begin{pmatrix} 0 & 2 & 0.5 & 0 \\ 2 & 0 & 1 & 4 \\ 0.5 & 1 & 0 & 0 \\ 0 & 4 & 0 & 0 \end{pmatrix}$$

$$A_{ii} = 0$$

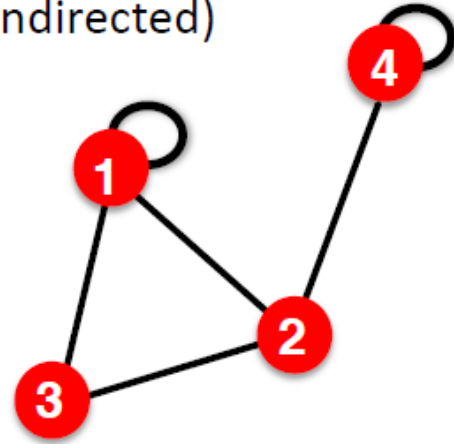
$$A_{ij} = A_{ji}$$

$$E = \frac{1}{2} \sum_{i,j=1}^N \text{nonzero}(A_{ij}) \quad \bar{k} = \frac{2E}{N}$$

More Graph Types

■ Self-edges (self-loops)

(undirected)



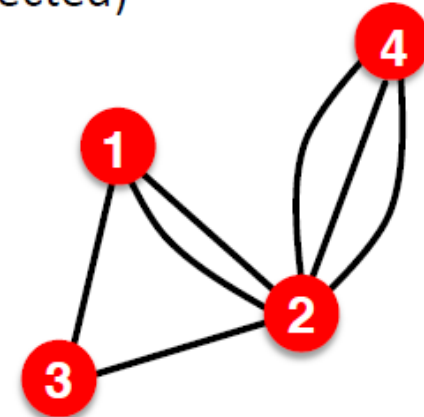
$$A_{ij} = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

$$E = \frac{1}{2} \sum_{i,j=1, i \neq j}^N A_{ij} + \sum_{i=1}^N A_{ii} \quad A_{ij} = A_{ji} \quad A_{ii} \neq 0$$

Examples: Proteins, Hyperlinks

■ Multigraph

(undirected)



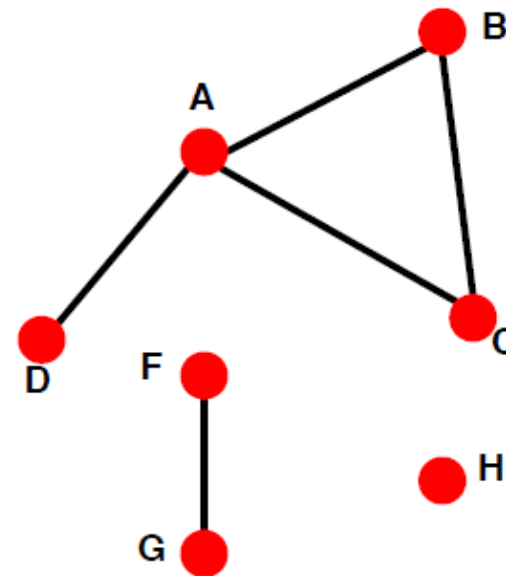
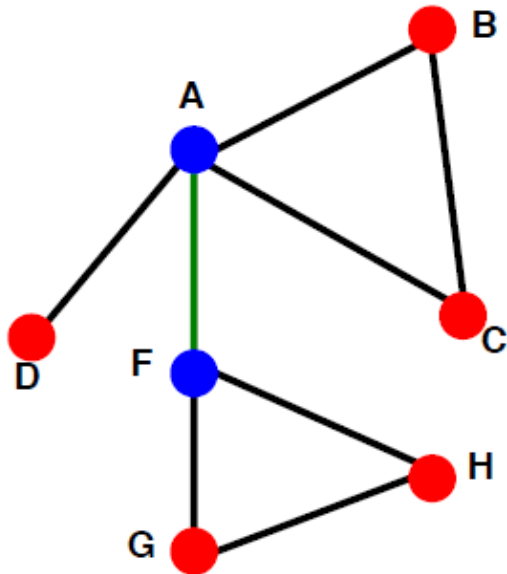
$$A_{ij} = \begin{pmatrix} 0 & 2 & 1 & 0 \\ 2 & 0 & 1 & 3 \\ 1 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 \end{pmatrix}$$

$$E = \frac{1}{2} \sum_{i,j=1}^N \text{nonzero}(A_{ij}) \quad A_{ij} = A_{ji} \quad A_{ii} = 0 \quad \bar{k} = \frac{2E}{N}$$

Examples: Communication, Collaboration

Connectivity of Undirected Graphs

- **Connected (undirected) graph:**
 - Any two vertices can be joined by a **path**
- A disconnected graph is made up by two or more connected components



Largest Component:
Giant Component

Isolated node (node H)

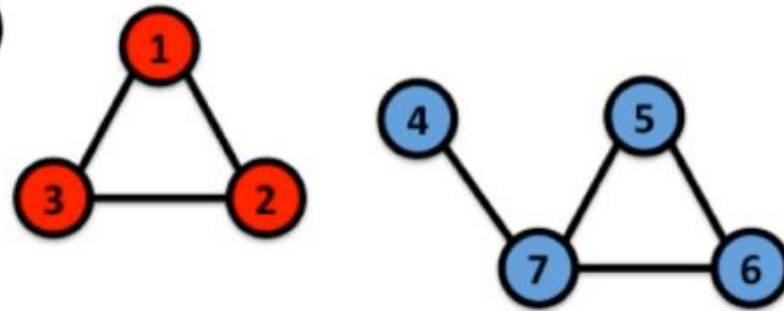
Bridge edge: If we erase the **edge**, the graph becomes disconnected

Connectivity

- The adjacency matrix of a network with several components can be written in a block-diagonal form, so that nonzero elements are confined to squares, with all other elements being zero:

Disconnected

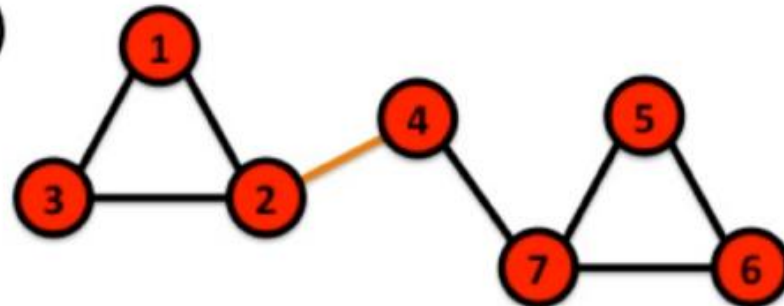
(a)



$$\begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}$$

Connected

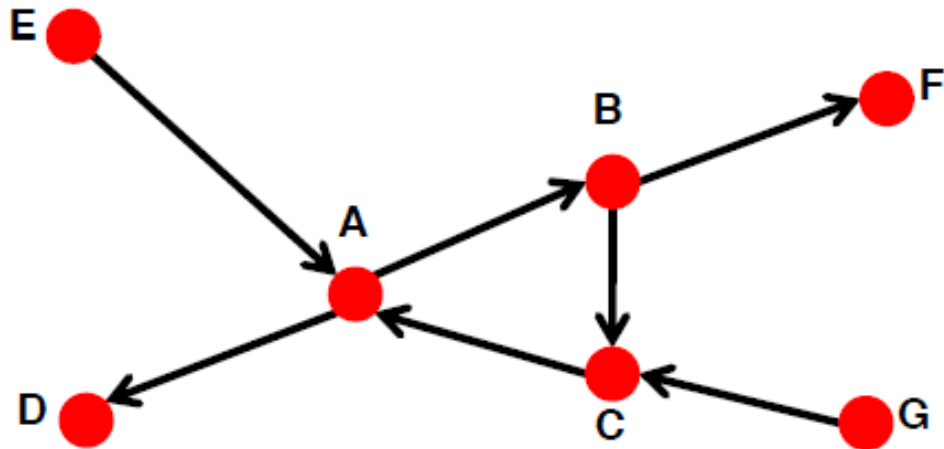
(b)



$$\begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}$$

Connectivity of Directed Graphs

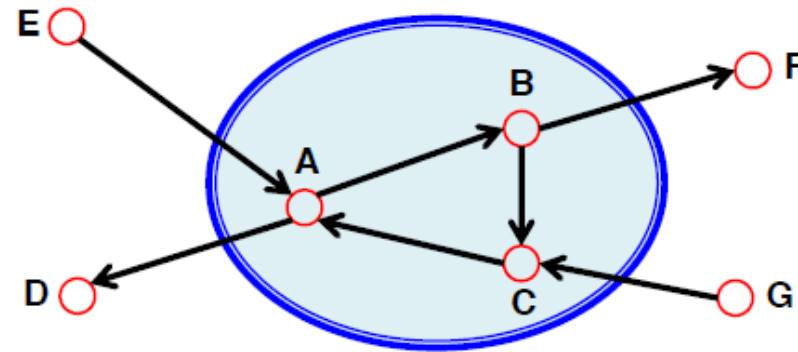
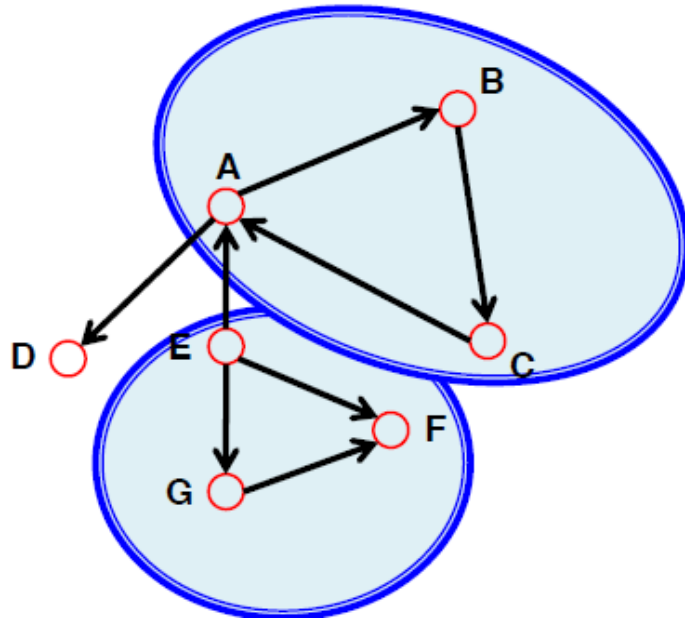
- **Strongly connected directed graph**
 - has a path from each node to every other node
- **Weakly connected directed graph**
 - is connected if we disregard the edge directions



Graph is connected but not strongly connected -
Can not get from F to G following edge directions

Connectivity of Directed Graphs

- **Strongly connected components** (SCCs) can be identified, but not every node is part of a nontrivial strongly connected component.



In-component: nodes that can reach the SCC,

Out-component: nodes that can be reached from the SCC.

who is in/out comp.?

Recap

- **Network representations by:**

- (un)directed , (un)weighted, multigraph, self-edge

Email network >> directed multigraph with self-edges

Facebook friendships >> undirected, unweighted

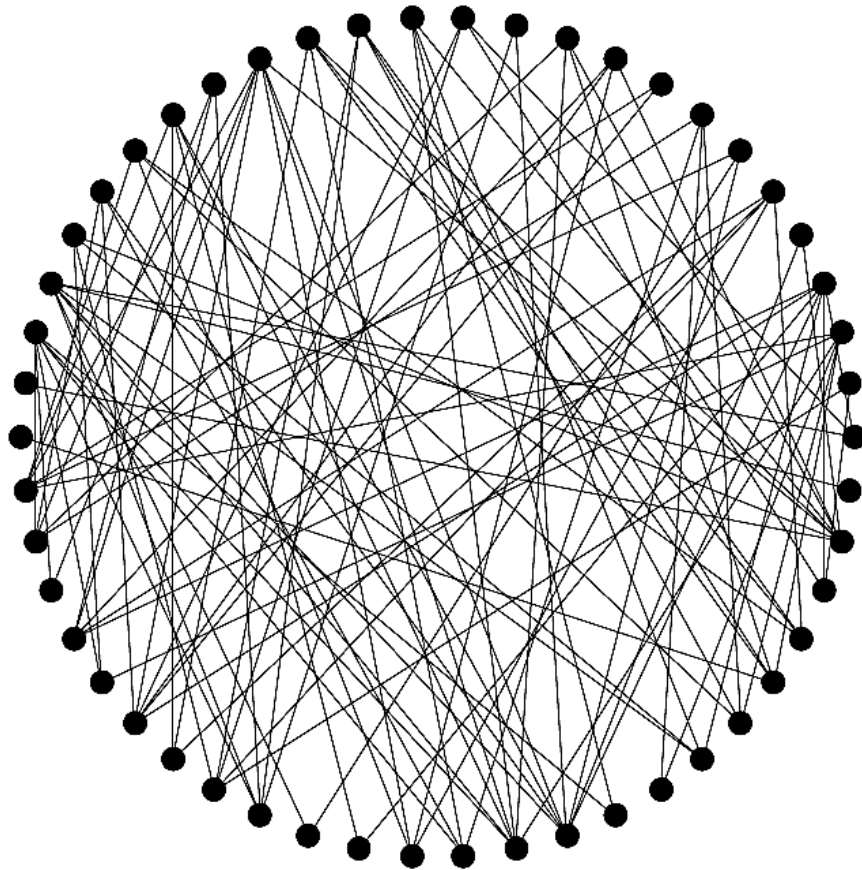
Citation networks >> unweighted, directed

Collaboration networks >> undirected multigraph or weighted graph

Mobile phone calls >> directed, (weighted?) multigraph

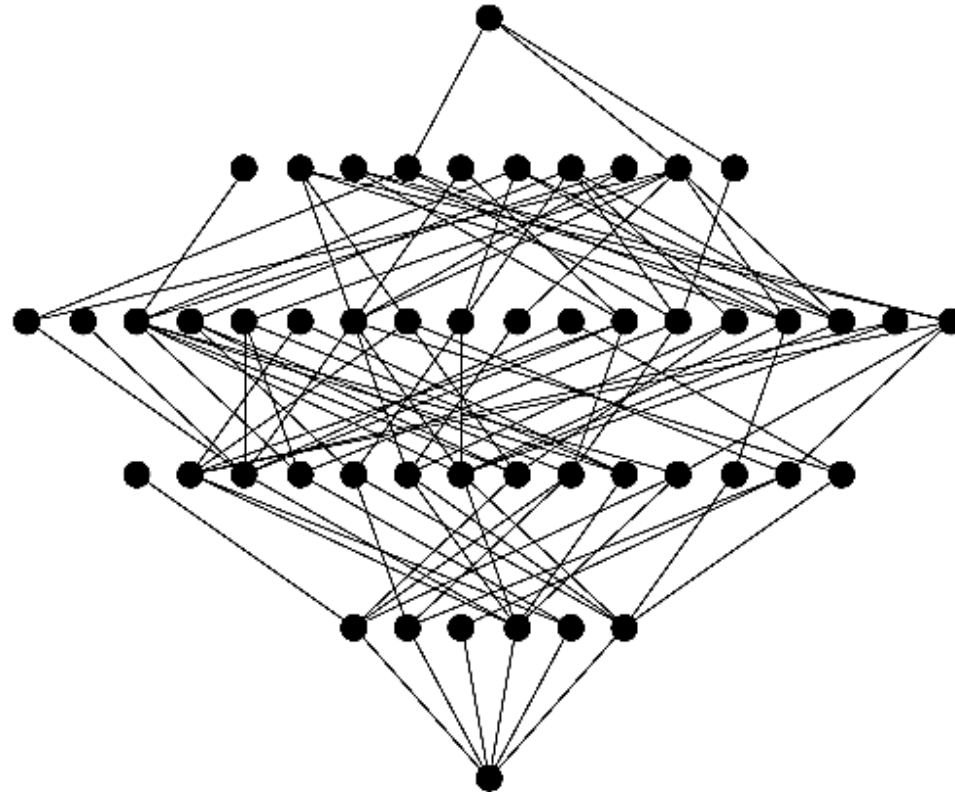
Graph Drawing (embedding)

- Circular embedding



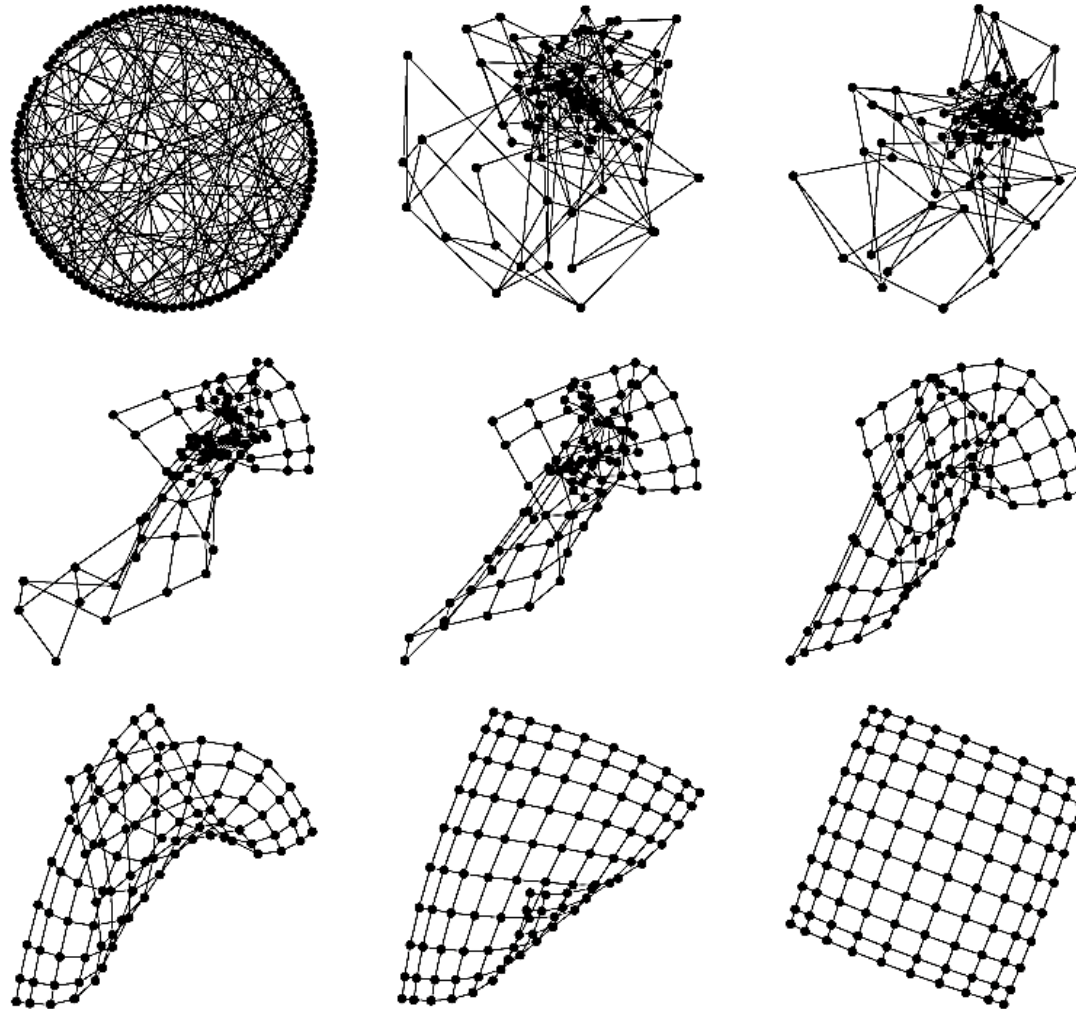
Graph Drawing

- Ranked embedding



Graph Drawing

- Spring embedding



Planar graphs

- A graph is **planar** if there exists an embedding in the 2D plane such that no two edges cross.
- A **plane graph** is a drawing of a planar graph such that no two edges intersect.

